Storyline

Chapter 5:
The Laws of Motion

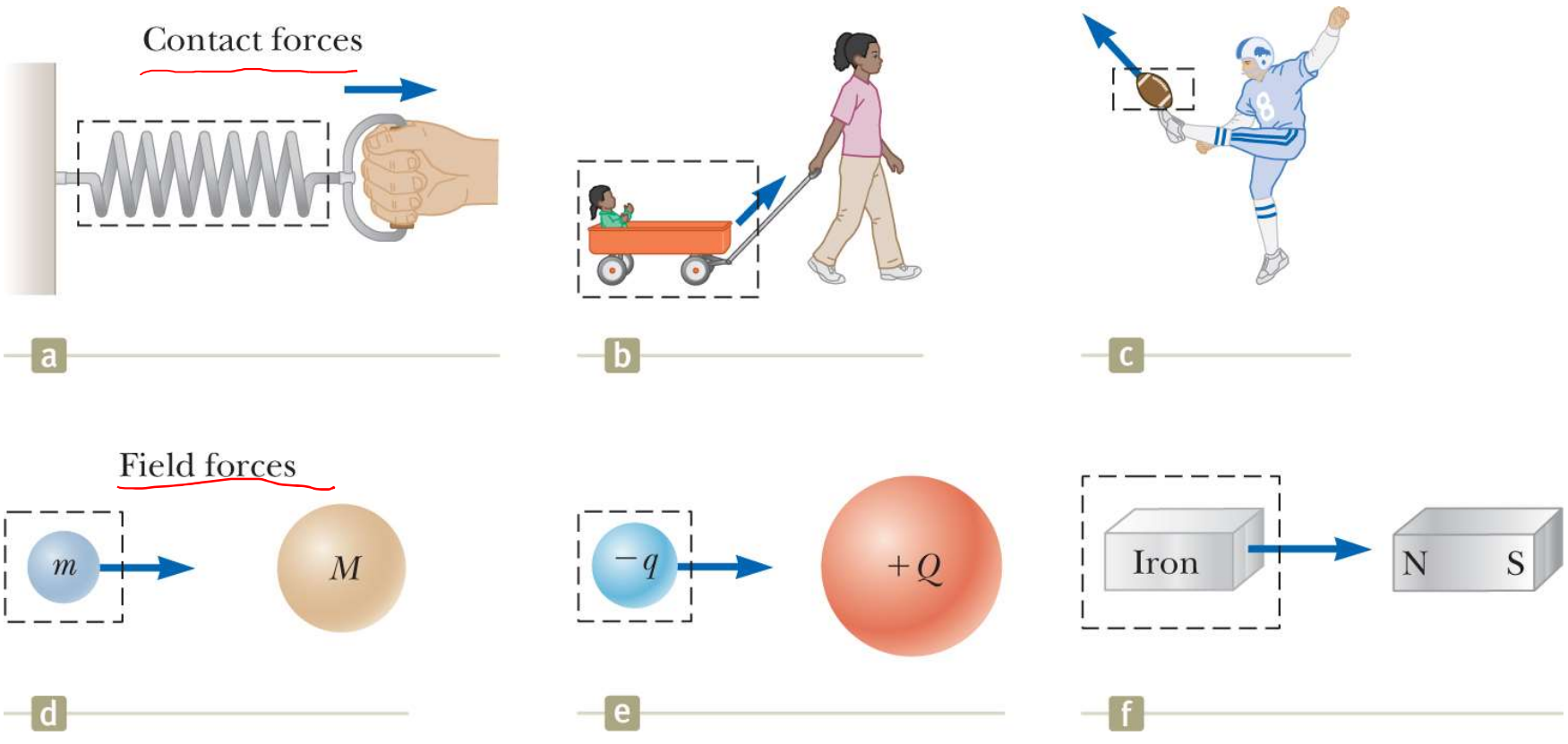


Physics for Scientists and Engineers, 10e
Raymond A. Serway
John W. Jewett, Jr.

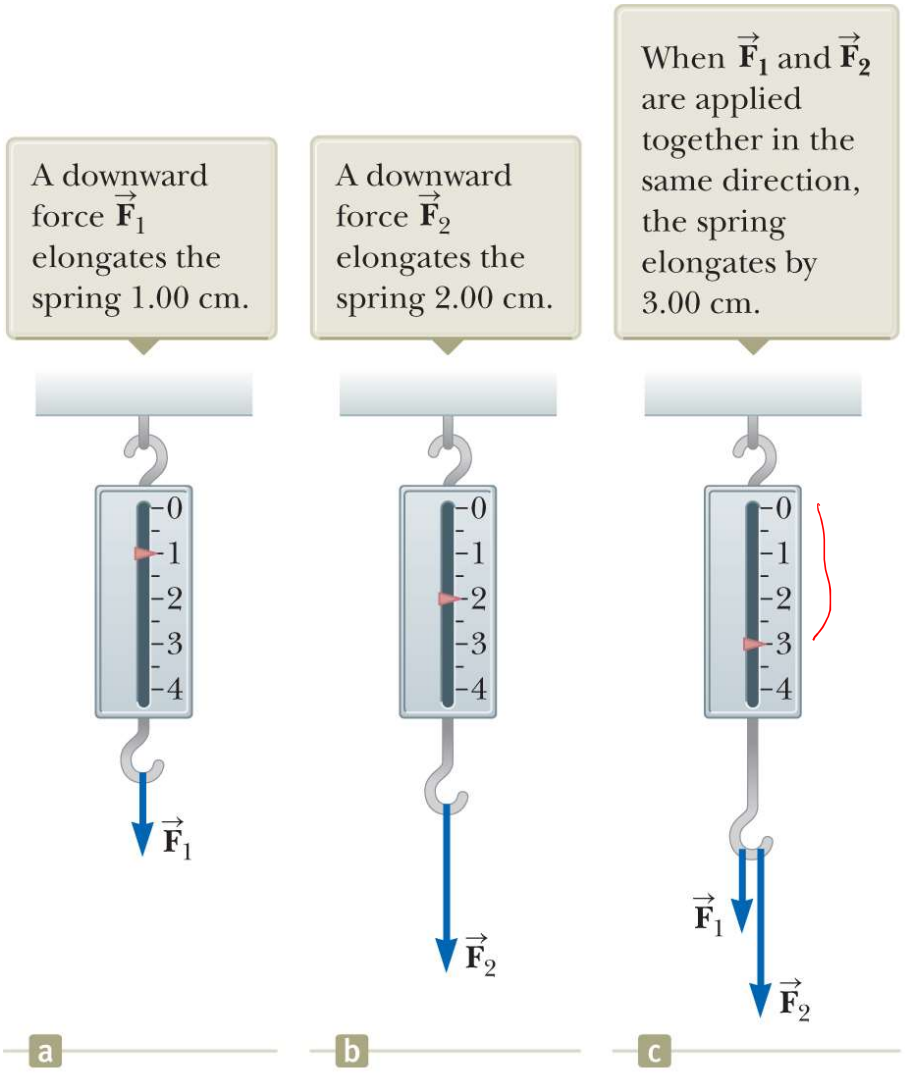
BROOKS/COLE
CENGAGE Learning

The Concept of Force

push
pull



The Concept of Force



The Concept of Force

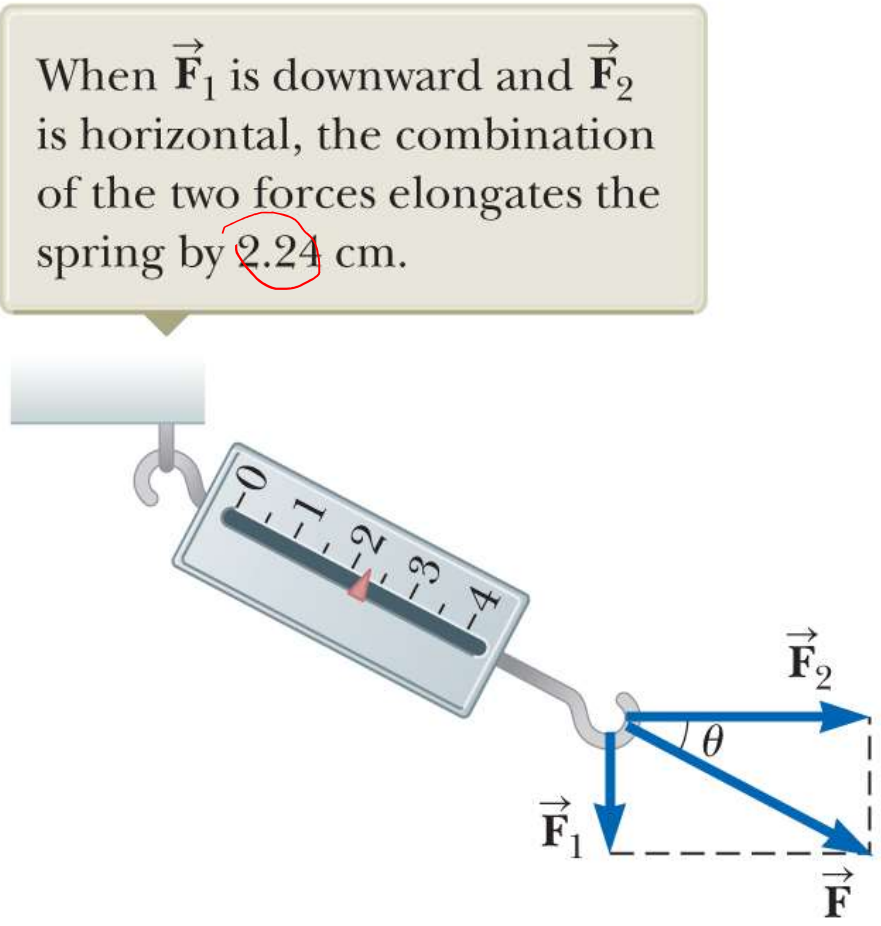
$$\left|\vec{\mathbf{F}}_1\right| = \sqrt{F_1^{\,2} + F_2^{\,2}}$$

$$= 2.24 \text{ units}$$

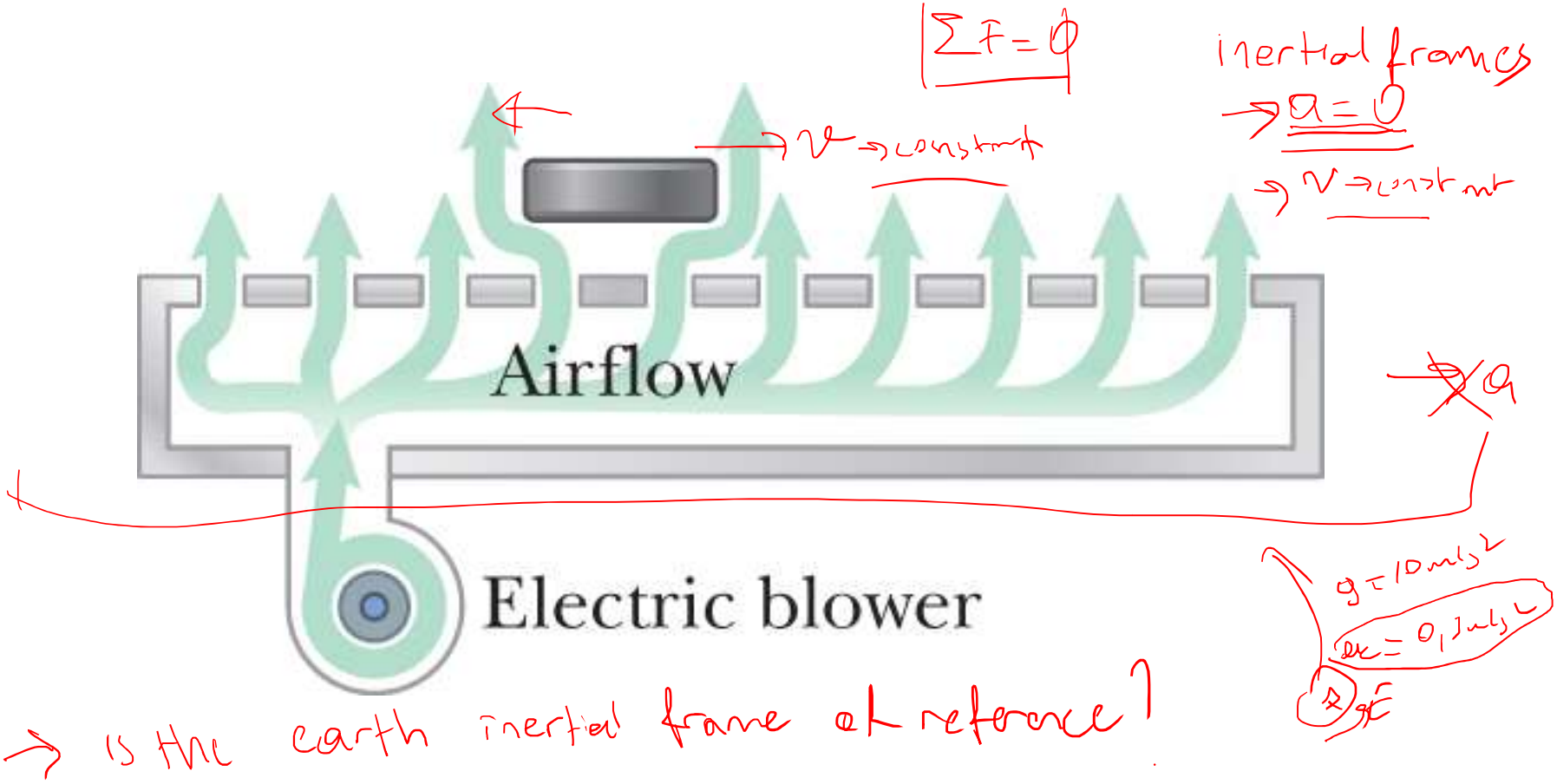
$$\theta = \tan^{-1}\left(\frac{1.00 \text{ unit}}{2.00 \text{ unit}}\right)$$

$$= \tan^{-1}(0.500)$$

$$= 26.6°$$



Newton's First Law and Inertial Frames



$\Sigma F = 0$

inertial frames

$a = 0$

$v \rightarrow$ constant

$v \rightarrow$ constant

$g = 10\ m/s^2$

$a_c = 0.1\ m/s^2$

→ is the earth inertial frame of reference?

Newton's First Law and Inertial Frames

If an object does not interact with other objects, it is possible to identify a reference frame in which the object has zero acceleration.

Inertial frame of reference

Newton's First Law and Inertial Frames



Newton's First Law and Inertial Frames

In the absence of external forces and when viewed from an inertial reference frame, an object at rest remains at rest and an object in motion continues in motion with a constant velocity (that is, with a constant speed in a straight line).

Quick Quiz 5.1

Which of the following statements is correct?

(a) It is possible for an object to have motion in the absence of forces on the object.

(b) It is possible to have forces on an object in the absence of motion of the object.

(c) Neither statement (a) nor statement (b) is correct.

(d) Both statements (a) and (b) are correct.

Quick Quiz 5.1

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Mass

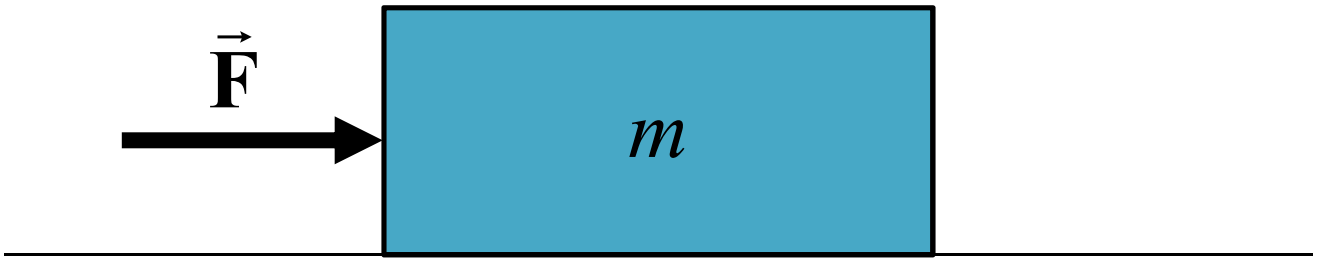
Mass is a property that specifies how much resistance an object exhibits to changes in its velocity

$$\frac{m_1}{m_2} \equiv \frac{a_2}{a_1}$$





Newton's Second Law



$$|\vec{\mathbf{a}}| \propto F \qquad |\vec{\mathbf{a}}| \propto \frac{1}{m}$$

Newton's Second Law

When viewed from an inertial reference frame, the acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass:

$$|\vec{\mathbf{a}}| \propto \frac{\sum \vec{\mathbf{F}}_{net}}{m_{total}}$$

Newton's Second Law

$$\sum \vec{\mathbf{F}}_{net} = m\vec{\mathbf{a}}$$

$$\sum F_x = ma_x \quad \sum F_y = ma_y \quad \sum F_z = ma_z$$

Quick Quiz 5.2

An object experiences no acceleration. Which of the following *cannot* be true for the object?

(a) A single force acts on the object.

(b) No forces act on the object.

(c) Forces act on the object, but the forces cancel.

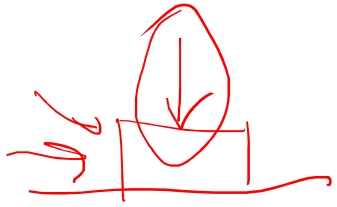
Quick Quiz 5.2

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(c) Forces act on the object, but the forces cancel.



Quick Quiz 5.3

You push an object, initially at rest, across a frictionless floor with a constant force for a time interval Δt, resulting in a final speed of v for the object. You then repeat the experiment, but with a force that is twice as large. What time interval is now required to reach the same final speed v?

(a) 4 Δt

(b) 2 Δt

(c) Δt

(d) Δt /2

(e) Δt /4

$v = a \Delta t$

$2a \cdot \frac{\Delta t}{2}$

$F = ma$

$\frac{F}{m} = a$

Quick Quiz 5.3

You push an object, initially at rest, across a frictionless floor with a constant force for a time interval Δt, resulting in a final speed of v for the object. You then repeat the experiment, but with a force that is twice as large. What time interval is now required to reach the same final speed v?

(a) $4\ \Delta t$

(b) $2\ \Delta t$

(c) Δt

(d) $\Delta t\ /2$

(e) $\Delta t\ /4$

Units of Force

SI unit of force: **newton** (N)

$$1\ \mathrm{N} \equiv 1\ \mathrm{kg} \cdot \mathrm{m/s^2}$$

$$1\ \mathrm{lb} \equiv 1\ \mathrm{slug} \cdot \mathrm{ft/s^2}$$

$$1\ \mathrm{N} \approx \frac{1}{4}\ \mathrm{lb}$$

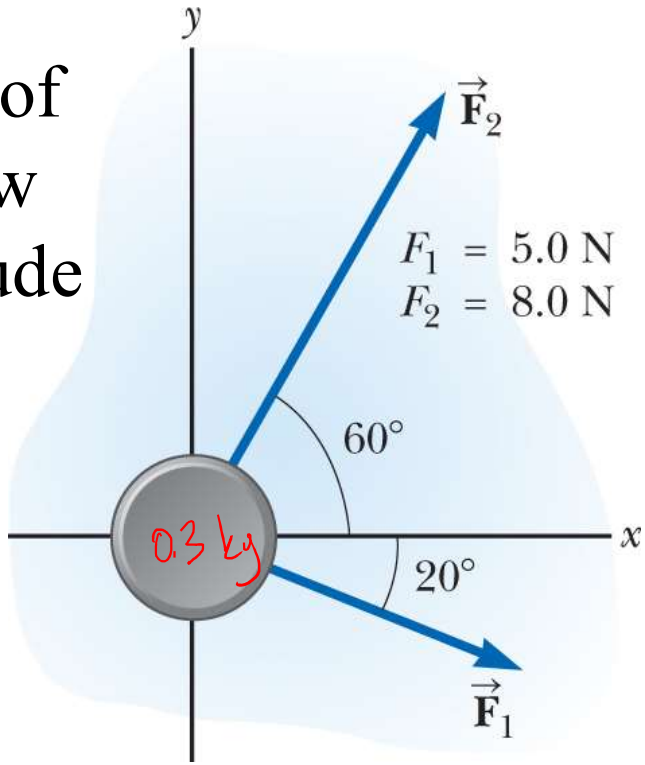
Newton's Second Law



$$\sum \vec{\mathbf{F}} = m\vec{\mathbf{a}}$$

Example 5.1:
An Accelerating Hockey Puck

A hockey puck having a mass of 0.30 kg slides on the frictionless, horizontal surface of an ice rink. Two hockey sticks strike the puck simultaneously, exerting the forces on the puck shown in the figure. The force $\mathbf{F}_1$ has a magnitude of 5.0 N, and is directed at $\theta = 20°$ below the x axis. The force $\mathbf{F}_2$ has a magnitude of 8.0 N and its direction is $\phi = 60°$ above the x axis. Determine both the magnitude and the direction of the puck's acceleration.



Example 5.1: An Accelerating Hockey Puck

$$\sum F_x = F_{1x} + F_{2x} = F_1 \cos\theta + F_2 \cos\phi$$

$$\sum F_y = F_{1y} + F_{2y} = F_1 \sin\theta + F_2 \sin\phi$$

$$\beta = \tan^{-1}\left(\frac{a_y}{a_x}\right)$$

$$a_x = \frac{\sum F_x}{m} = \frac{F_1 \cos\theta + F_2 \cos\phi}{m}$$

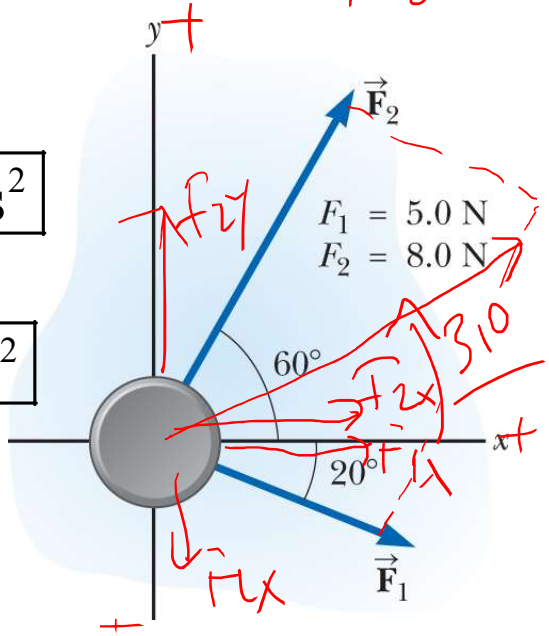
$$= \tan^{-1}\left(\frac{17}{29}\right) = \boxed{31^\circ}$$

$$a_y = \frac{\sum F_y}{m} = \frac{F_1 \sin\theta + F_2 \sin\phi}{m}$$

$$a_x = \frac{(5.0\text{ N})\cos(-20^\circ) + (8.0\text{ N})\cos(60^\circ)}{0.30\text{ kg}} = \boxed{29\text{ m/s}^2}$$

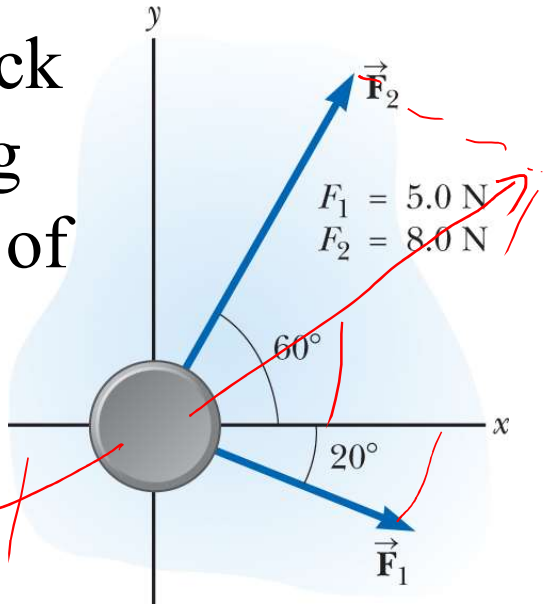
$$a_y = \frac{(5.0\text{ N})\sin(-20^\circ) + (8.0\text{ N})\sin(60^\circ)}{0.30\text{ kg}} = \boxed{17\text{ m/s}^2}$$

$$a = \sqrt{(29\text{ m/s}^2)^2 + (17\text{ m/s}^2)^2} = \boxed{34\text{ m/s}^2}$$



Example 5.1:
An Accelerating Hockey Puck

Suppose three hockey sticks strike the puck simultaneously, with two of them exerting the forces shown in the figure. The result of the three forces is that the hockey puck shows *no* acceleration. What must be the components of the third force?



$$F_{3x} = -\sum F_x = -(0.30\ \text{kg})(29\ \text{m/s}^2) = -8.7\ \text{N}$$

$$F_{3y} = -\sum F_x = -(0.30\ \text{kg})(17\ \text{m/s}^2) = -5.2\ \text{N}$$

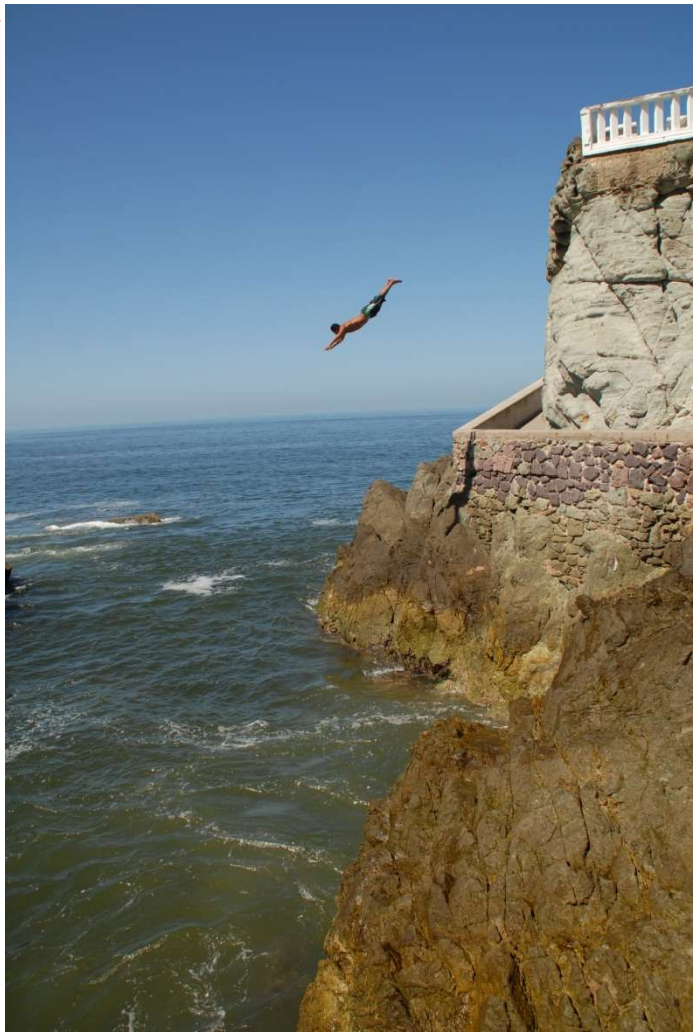
The Gravitational Force and Weight

$$\sum \vec{\mathbf{F}} = m\vec{\mathbf{a}}$$

with $\vec{\mathbf{a}} = \vec{\mathbf{g}}$ and $\sum \vec{\mathbf{F}} = \vec{\mathbf{F}}_g$

$$\vec{\mathbf{F}}_g = m\vec{\mathbf{g}}$$

$$F_g = mg$$



The Gravitational Force and Weight

g varies with geographic location

$$F_g = mg$$



The Gravitational Force and Weight

$$F_g = mg_{\text{Moon}}$$

$$g_{\text{Moon}} \approx 1.6 \text{ m/s}^2$$

inertial mass vs. gravitational mass



Quick Quiz 5.4

Suppose you are talking by interplanetary telephone to a friend who lives on the Moon. He tells you that he has just won a newton of gold in a contest. Excitedly, you tell him that you entered the Earth version of the same contest and also won a newton of gold! Who is richer?

(a) You are.

(b) Your friend is.

(c) You are equally rich.

Quick Quiz 5.4

Suppose you are talking by interplanetary telephone to a friend who lives on the Moon. He tells you that he has just won a newton of gold in a contest. Excitedly, you tell him that you entered the Earth version of the same contest and also won a newton of gold! Who is richer?

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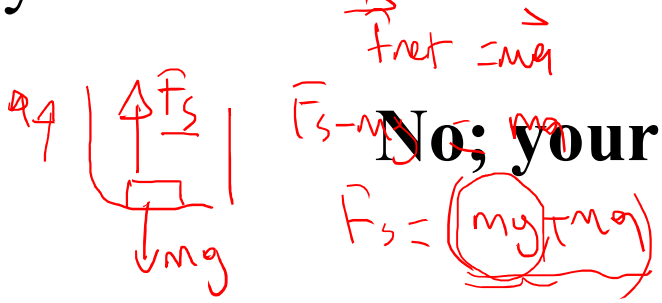
(c) You are equally rich.

$1 = 6m \quad \frac{g}{6}$

$1 = m\,g$

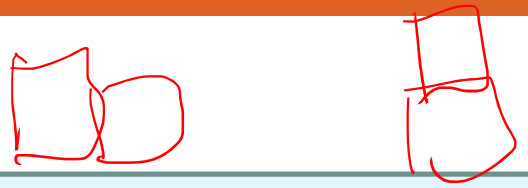
Conceptual Example 5.2: How Much Do You Weigh in an Elevator

You have most likely been in an elevator that accelerates upward as it moves toward a higher floor. In this case, you feel heavier. In fact, if you are standing on a bathroom scale at the time, the scale measures a force having a magnitude that is greater than your weight. Therefore, you have tactile and measured evidence that leads you to believe you are heavier in this situation. *Are* you heavier?



No; your weight is unchanged.

Newton's Third Law

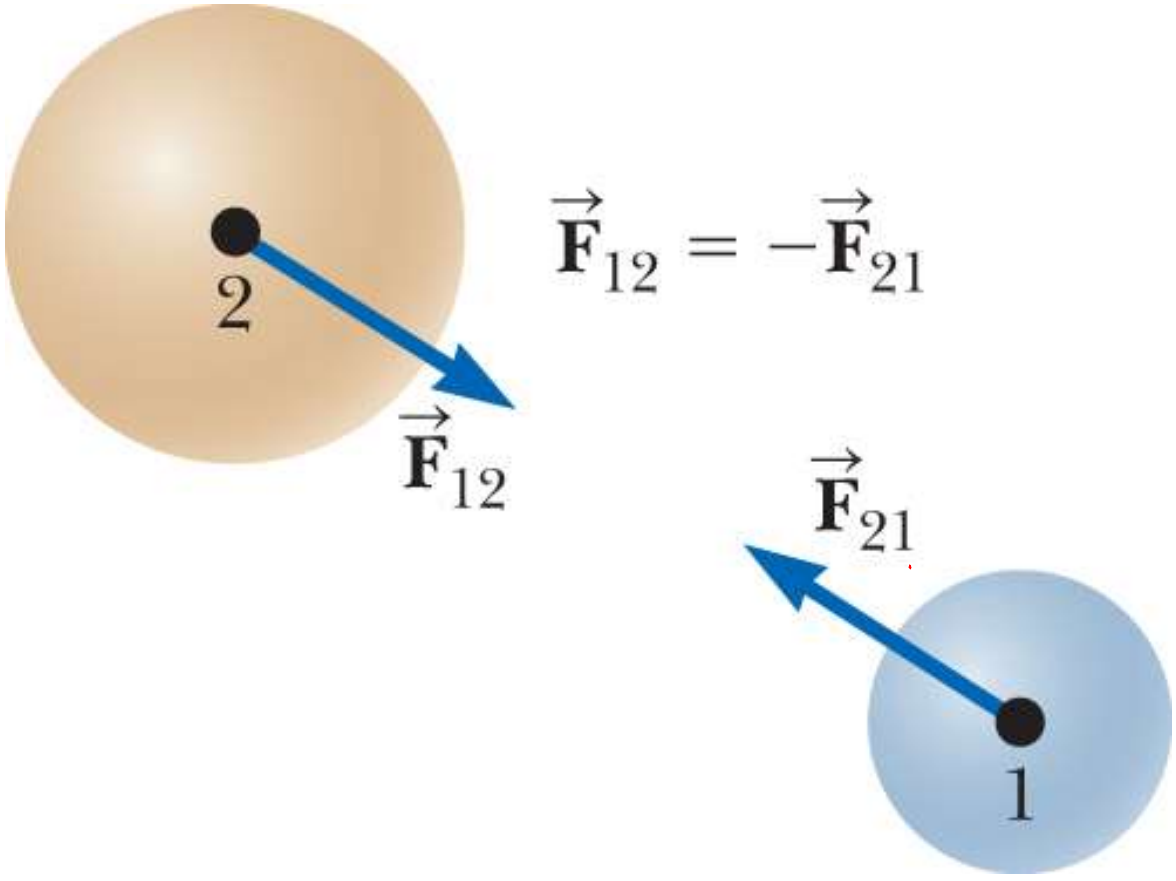


If two objects interact, the force $\vec{\mathbf{F}}_{12}$ exerted by object 1 on object 2 is equal in magnitude and opposite in direction to the force $\vec{\mathbf{F}}_{21}$ exerted by object 2 on object 1:

$$\vec{\mathbf{F}}_{12} = -\vec{\mathbf{F}}_{21}$$

$$\vec{F}_{12} + \vec{F}_{21} = 0 = m(0)$$

Newton's Third Law

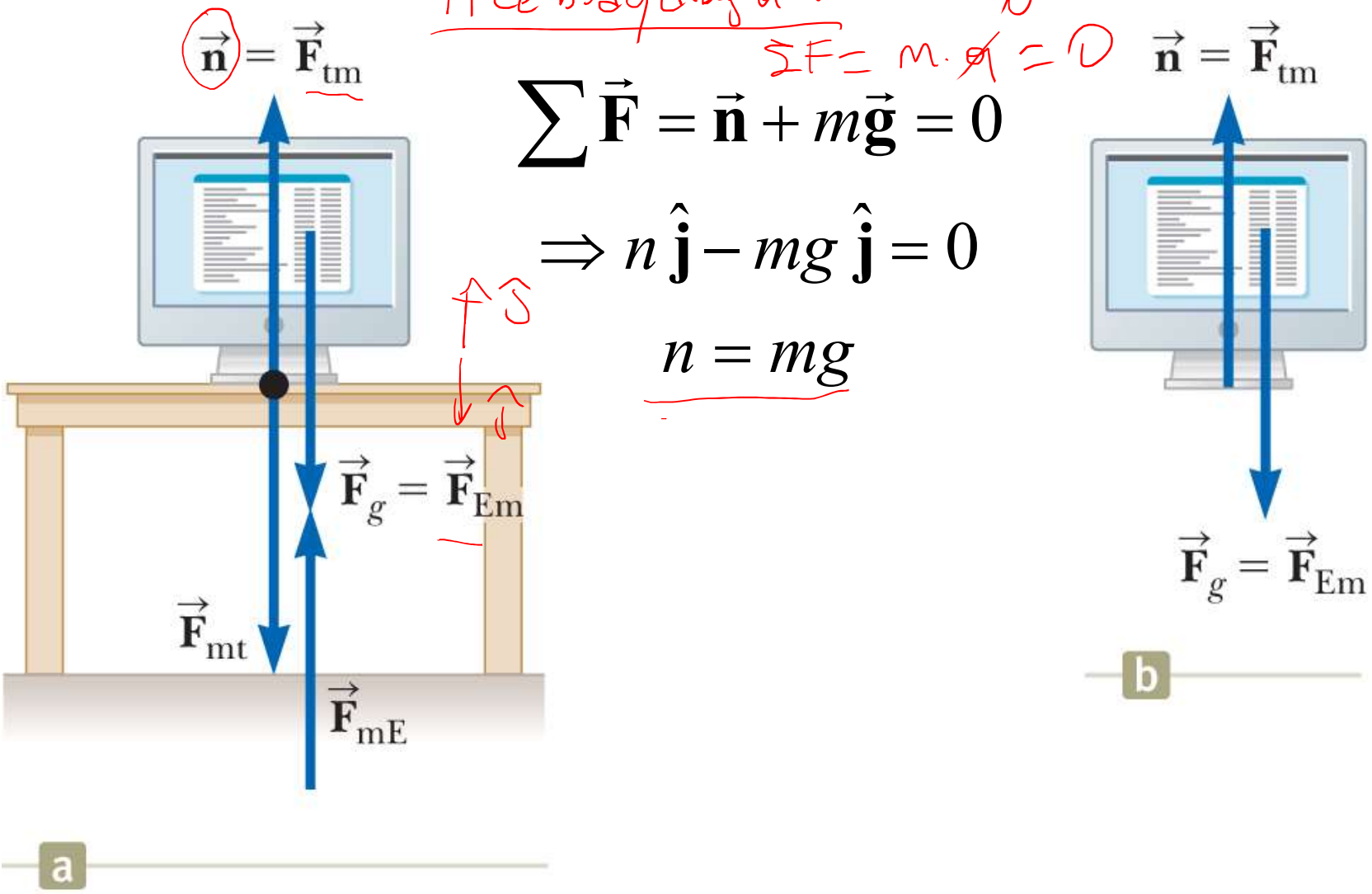


$$\vec{F}_{12} = -\vec{F}_{21}$$

$$\vec{F}_{g} = \vec{F}_{Ep}$$

$$\vec{F}_{pE} = -\vec{F}_{Ep}$$

Newton's Third Law



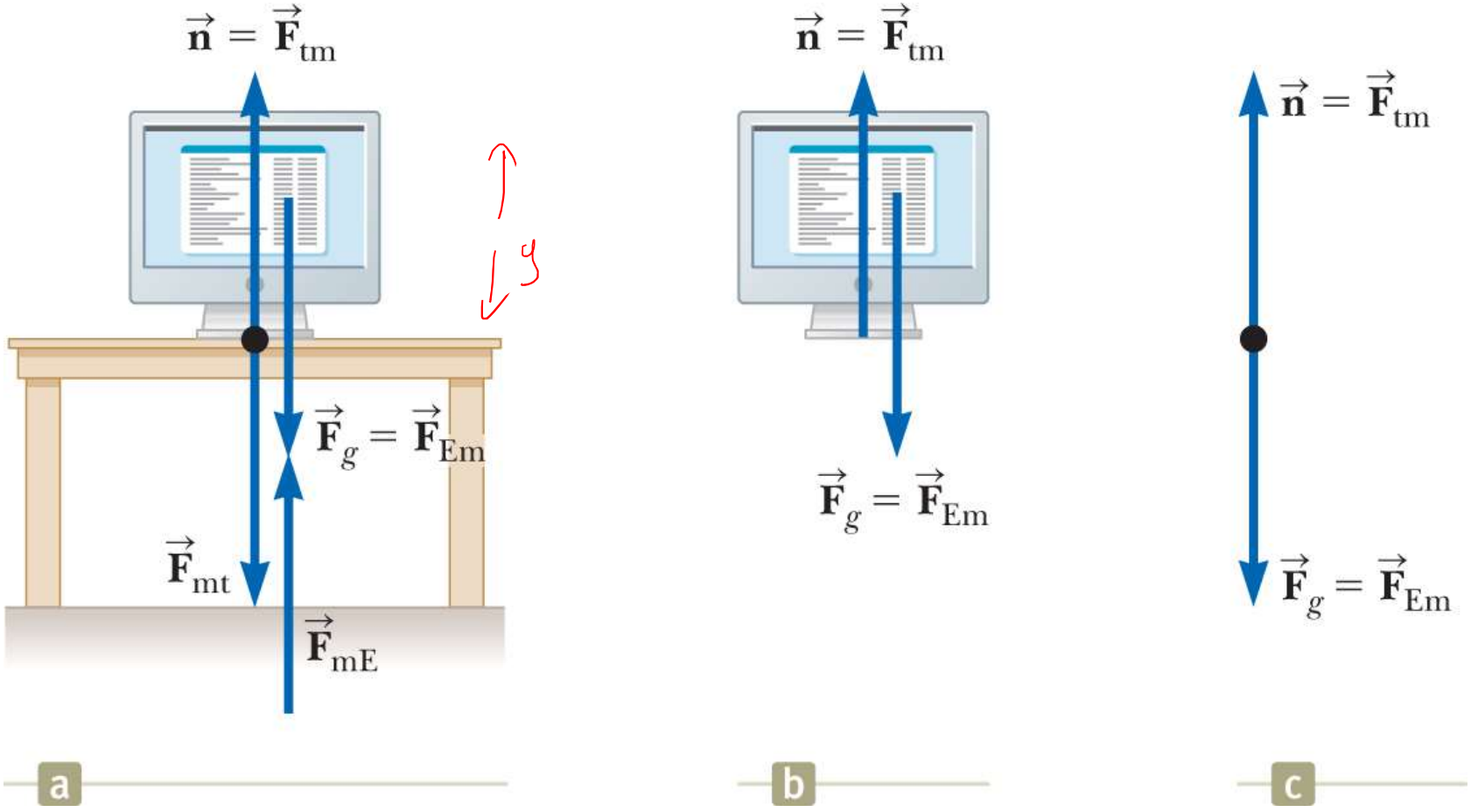
Free body diagram

$$\sum \vec{\mathbf{F}} = \vec{\mathbf{n}} + m\vec{\mathbf{g}} = 0$$

$$\Rightarrow n\,\hat{\mathbf{j}} - mg\,\hat{\mathbf{j}} = 0$$

$$n = mg$$

Free-Body Diagram



Quick Quiz 5.5 Part I

If a fly collides with the windshield of a fast-moving bus, which experiences an impact force with a larger magnitude?

(a) the fly

(b) the bus

(c) The same force is experienced by both.

Quick Quiz 5.5 Part I

If a fly collides with the windshield of a fast-moving bus, which experiences an impact force with a larger magnitude?

(a) the fly

(b) the bus

(c) The same force is experienced by both.

Quick Quiz 5.5 Part II

If a fly collides with the windshield of a fast-moving bus, which experiences the greater acceleration?

(a) the fly

(b) the bus

(c) The same acceleration is experienced by both.



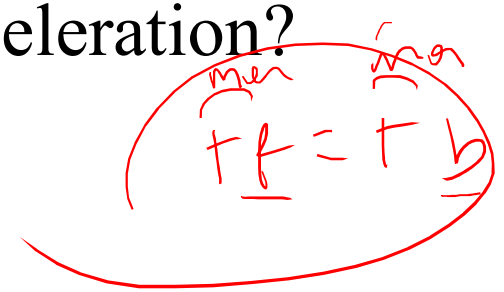
Quick Quiz 5.5 Part II

If a fly collides with the windshield of a fast-moving bus, which experiences the greater acceleration?

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(c) The same acceleration is experienced by both.



Conceptual Example 5.3: You Push Me and I'll Push You

A large man and a small boy stand facing each other on frictionless ice. They put their hands together and push against each other so that they move apart. Who moves away with the higher speed?

The boy



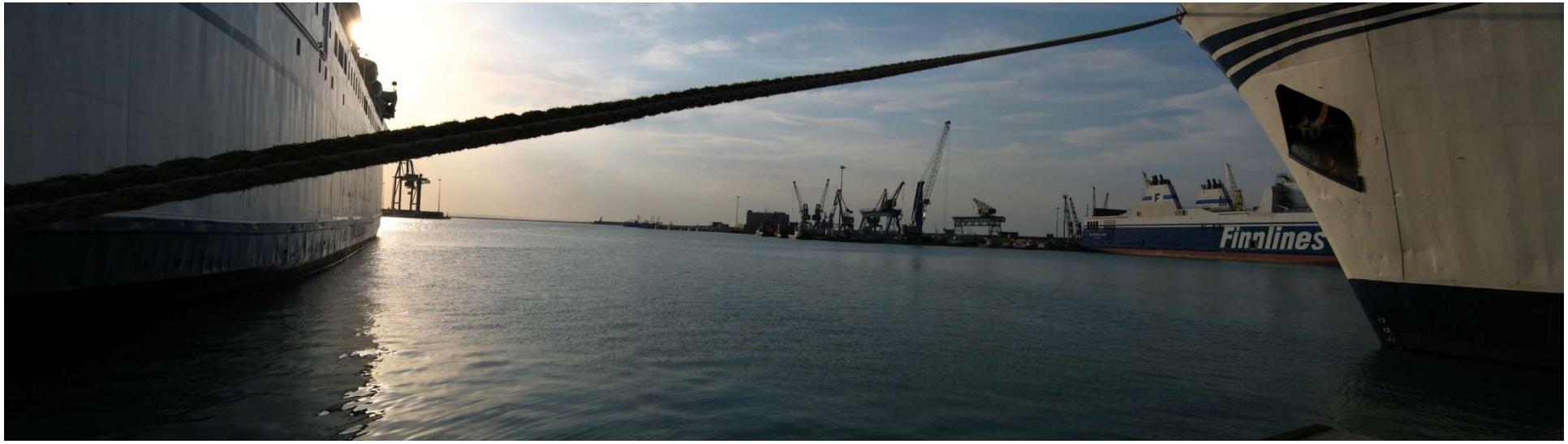
Conceptual Example 5.3: You Push Me and I'll Push You

Who moves farther while their hands are in contact?

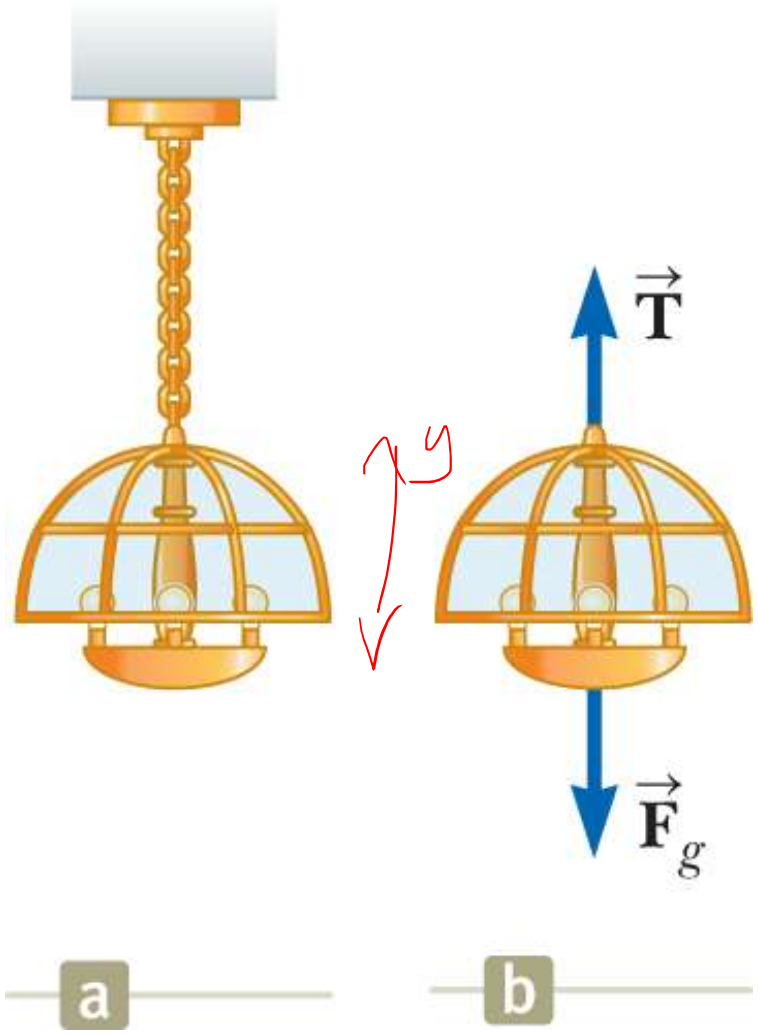
The boy

Analysis Models
Using Newton's Second Law

$$\sum \vec{F}_{net} = m\vec{a}$$



Analysis Model: The Particle in Equilibrium



$$\sum \vec{\mathbf{F}} = 0$$

$$\sum \mathbf{F}_y = T - F_g = 0$$

$$\text{or } T = F_g$$

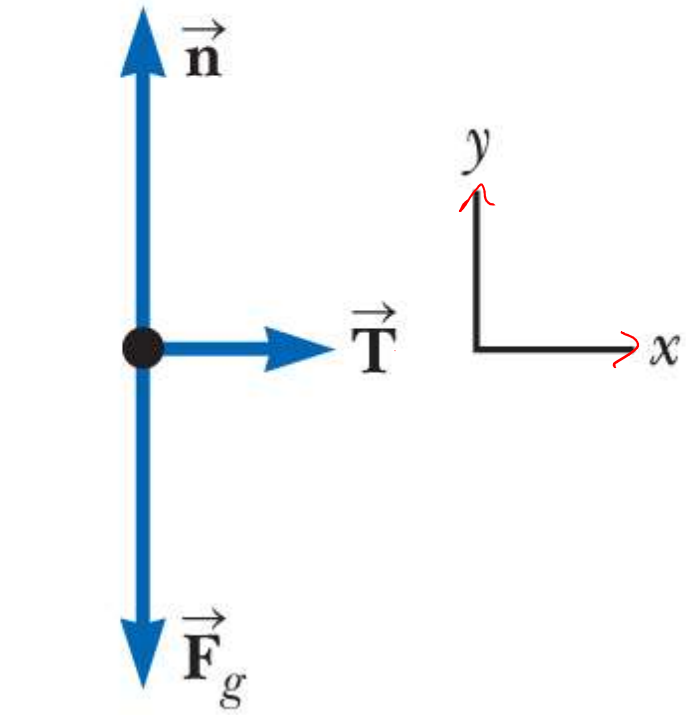
$T = mg$

Analysis Model:
The Particle Under a Net Force

$$\sum \vec{\mathbf{F}} = m\vec{\mathbf{a}}$$



a

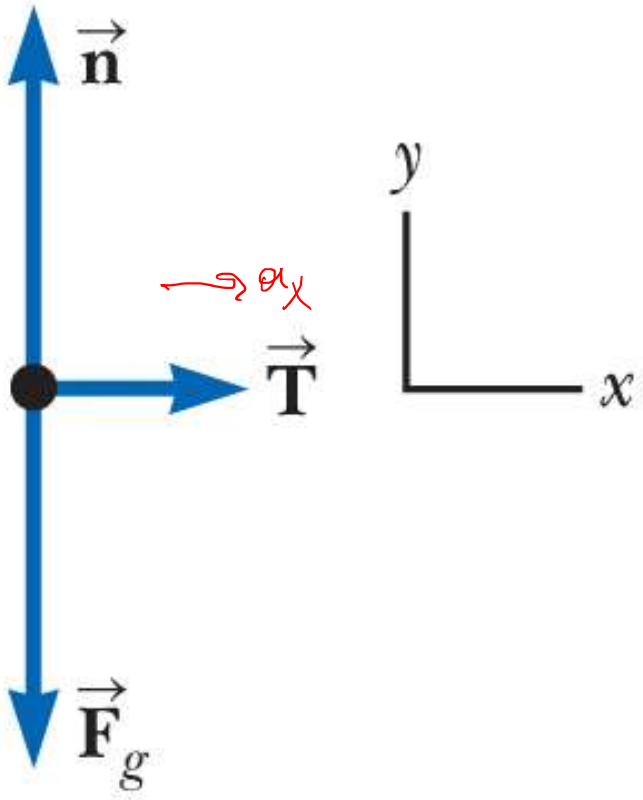


b

Analysis Model:
The Particle Under a Net Force

$$\sum F_x = T = ma_x \quad \text{or} \quad a_x = \frac{T}{m}$$

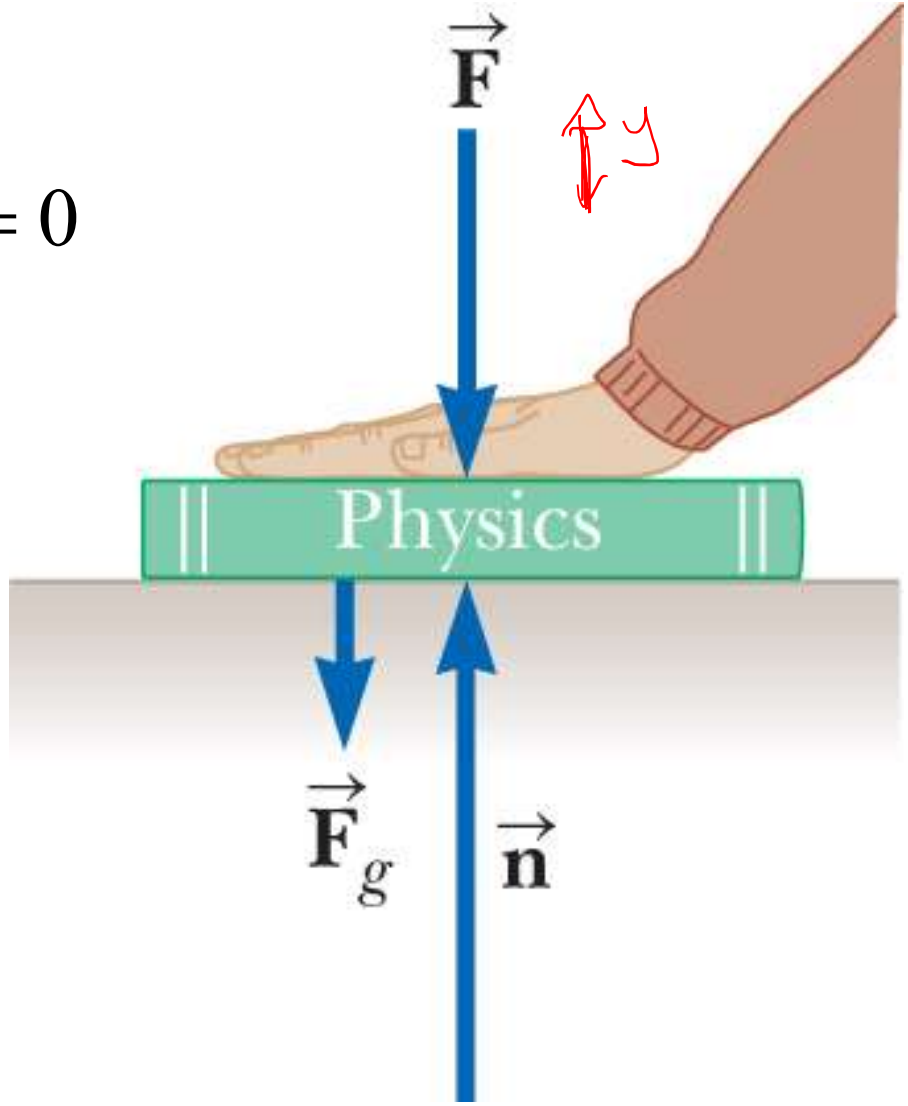
$$\sum F_y = n - F_g = 0 \quad \text{or} \quad n = F_g$$



Analysis Model:
The Particle Under a Net Force

$$\sum F_y = 0 \Rightarrow n - F_g - F = 0$$

$$n = F_g + F = mg + F$$



Analysis Model: Particle in Equilibrium

$$\sum \vec{\mathbf{F}} = 0$$

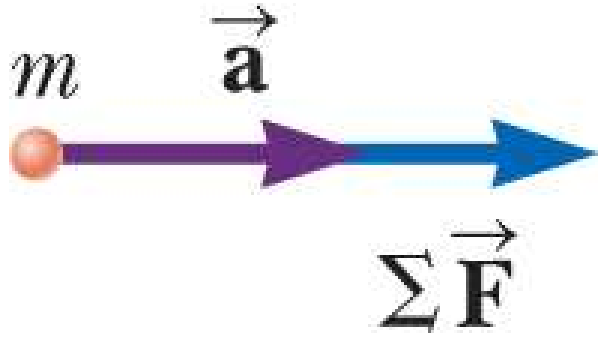
$$\vec{\mathbf{a}} = 0$$



$$\Sigma \vec{\mathbf{F}} = 0$$

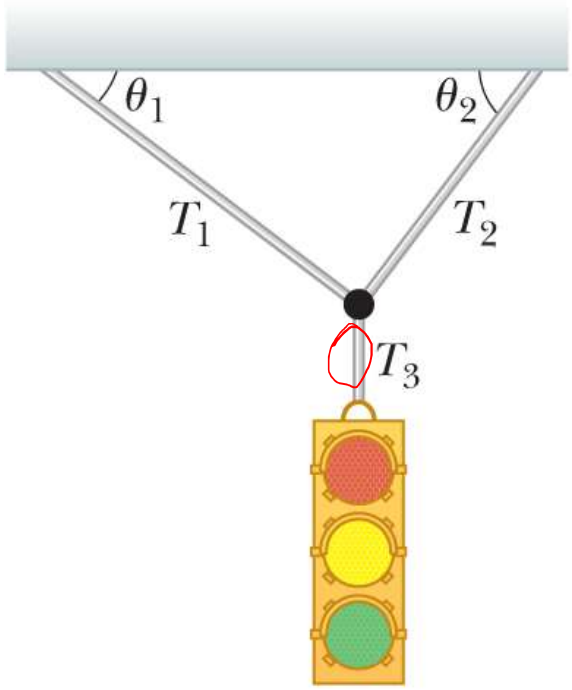
Analysis Model:
Particle Under a Net Force

$$\sum \vec{\mathbf{F}} = m\vec{\mathbf{a}}$$



Example 5.4:
A Traffic Light at Rest

A traffic light weighing 122 N hangs from a cable tied to two other cables fastened to a support as the figure. The upper cables make angles of $\theta_1 = 37.0°$ and $\theta_2 = 53.0°$ with the horizontal. These upper cables are not as strong as the vertical cable and will break if the tension in them exceeds 100 N. Does the traffic light remain hanging in this situation, or will one of the cables break?



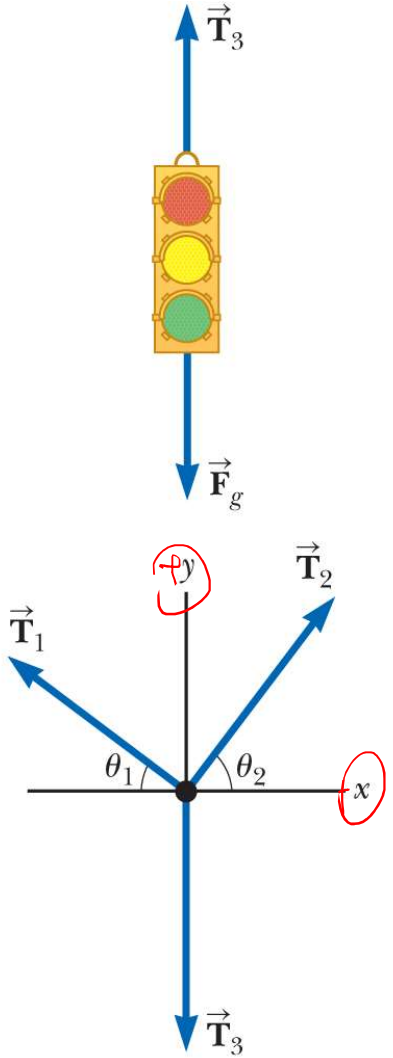
Example 5.4:
A Traffic Light at Rest

$$\sum F_y = 0 \Rightarrow T_3 - F_g = 0 \Rightarrow T_3 = F_g$$

Force	x Component	y Component
$\vec{\mathbf{T}}_1$	$-T_1 \cos\theta_1$	$T_1 \sin\theta_1$
$\vec{\mathbf{T}}_2$	$T_2 \cos\theta_2$	$T_2 \sin\theta_2$
$\vec{\mathbf{T}}_3$	0	$-F_g$

$$(1)\ \sum F_x = -T_1 \cos\theta_1 + T_2 \cos\theta_2 = 0$$

$$(2)\ \sum F_y = T_1 \sin\theta_1 + T_2 \sin\theta_2 + \left(-F_g\right) = 0$$



Example 5.4:
A Traffic Light at Rest

$$-T_1 \cos\theta_1 + T_2 \cos\theta_2 = 0 \Rightarrow T_2 = T_1\left(\frac{\cos\theta_1}{\cos\theta_2}\right)$$

$$T_1 \sin\theta_1 + T_2 \sin\theta_2 + \left(-F_g\right) = 0 \Rightarrow T_1 \sin\theta_1 + T_1\left(\frac{\cos\theta_1}{\cos\theta_2}\right)\left(\sin\theta_2\right) - F_g = 0$$

$$T_1 = \frac{F_g}{\sin\theta_1 + \cos\theta_1 \tan\theta_2}$$

$$T_1 = \frac{122\ \text{N}}{\sin 37.0^\circ + \cos 37.0^\circ \tan 53.0^\circ} = \boxed{73.4\ \text{N}} < 100\text{N}$$

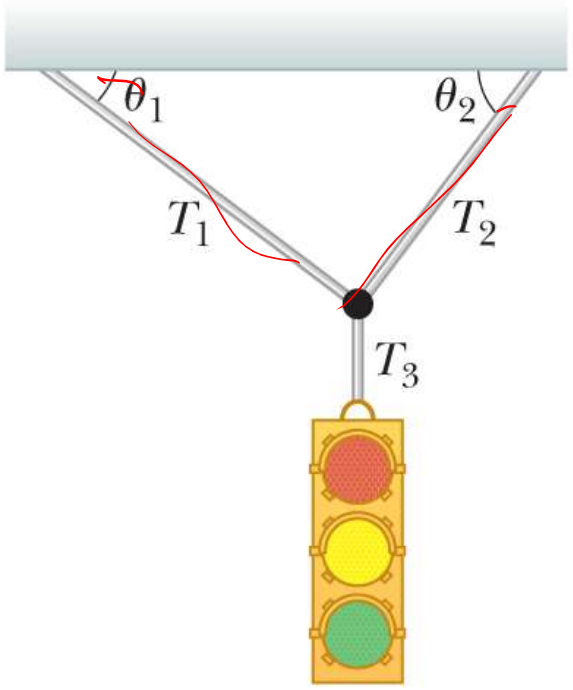
$$T_2 = \left(73.4\ \text{N}\right)\left(\frac{\cos 37.0^\circ}{\cos 53.0^\circ}\right) = \boxed{97.4\ \text{N}} < 100$$

Example 5.4:
A Traffic Light at Rest

Suppose the two angles in the figure are equal. What would be the relationship between T_1 and T_2?

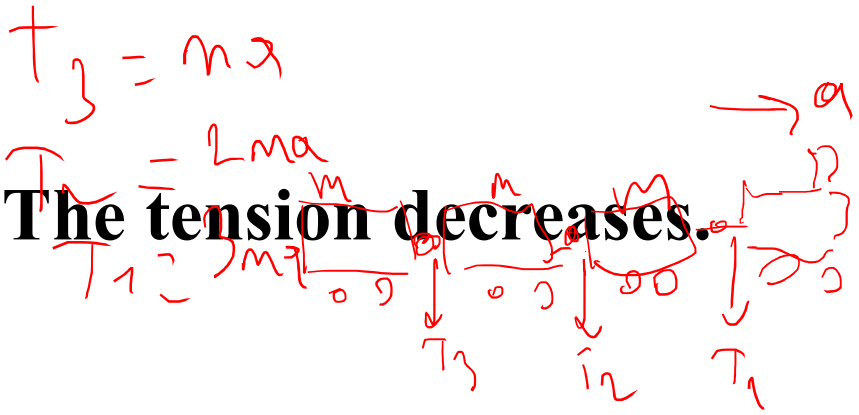
$\theta_1 = \theta_2$

$$T_2 = T_1\left(\frac{\cos\theta_1}{\cos\theta_2}\right) = T_1$$



Conceptual Example 5.5: Forces Between Cars in a Train

Train cars are connected by *couplers,* which are under tension as the locomotive pulls the train. Imagine you are on a train speeding up with a constant acceleration. As you move through the train from the locomotive to the last car, measuring the tension in each set of couplers, does the tension increase, decrease, or stay the same?



The tension decreases.

Conceptual Example 5.5: Forces Between Cars in a Train

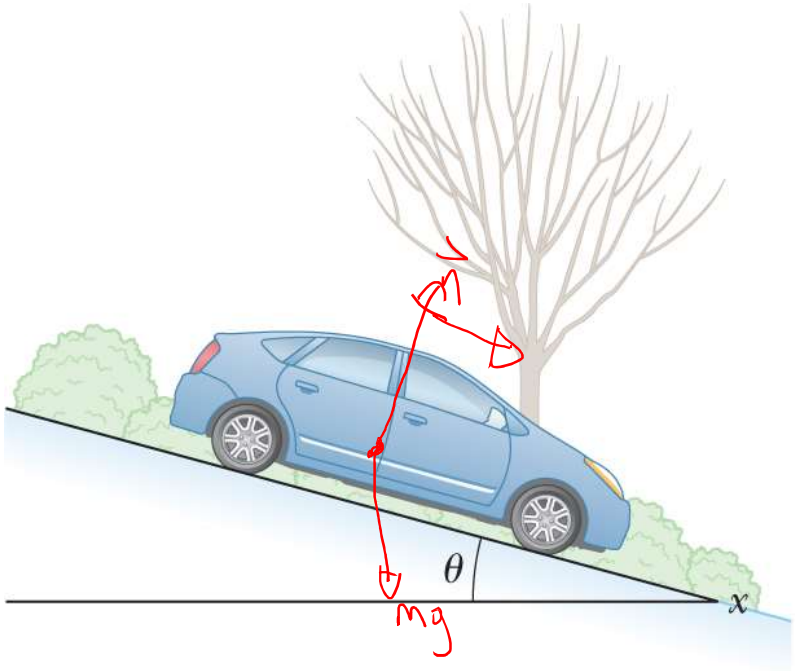
When the engineer applies the brakes, the couplers are under compression. How does this compression force vary from the locomotive to the last car? (Assume only the brakes on the wheels of the engine are applied.)

The force decreases.

Example 5.6:
The Runaway Car

A car of mass m is on an icy driveway inclined at an angle θ as in the figure.

(A) Find the acceleration of the car, assuming the driveway is frictionless.

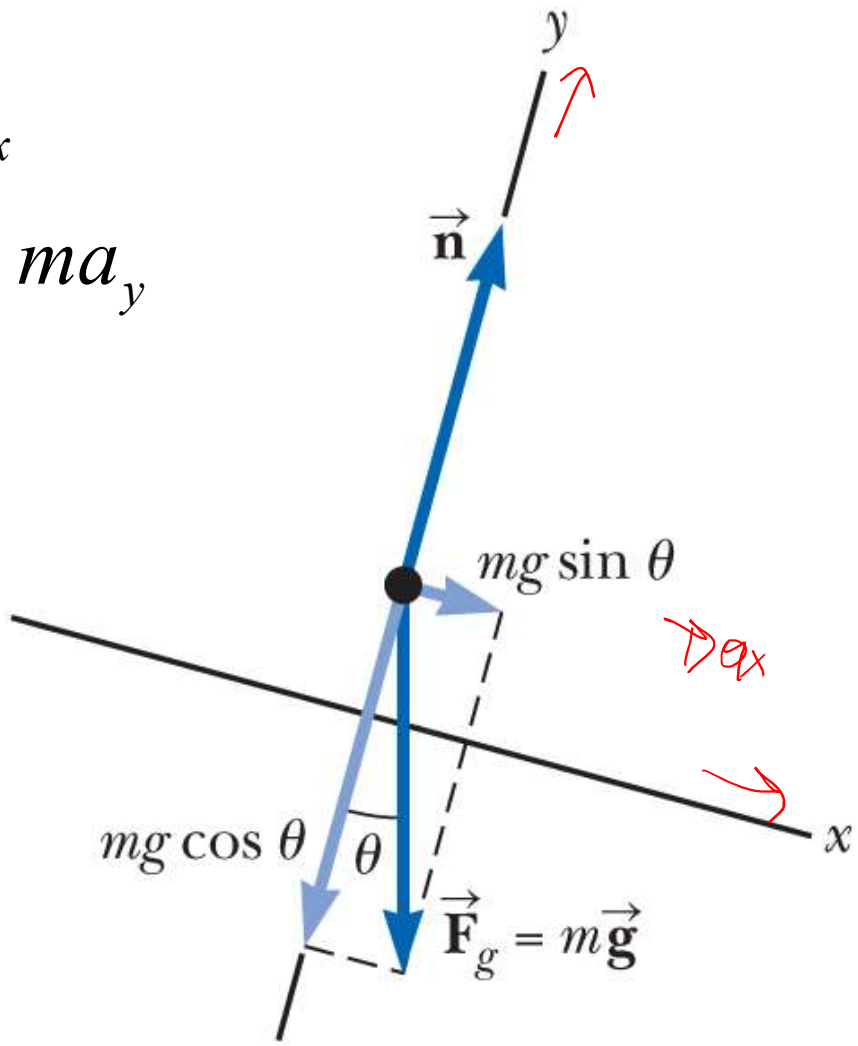


Example 5.6:
The Runaway Car

$$\sum F_x = mg\sin\theta = ma_x$$

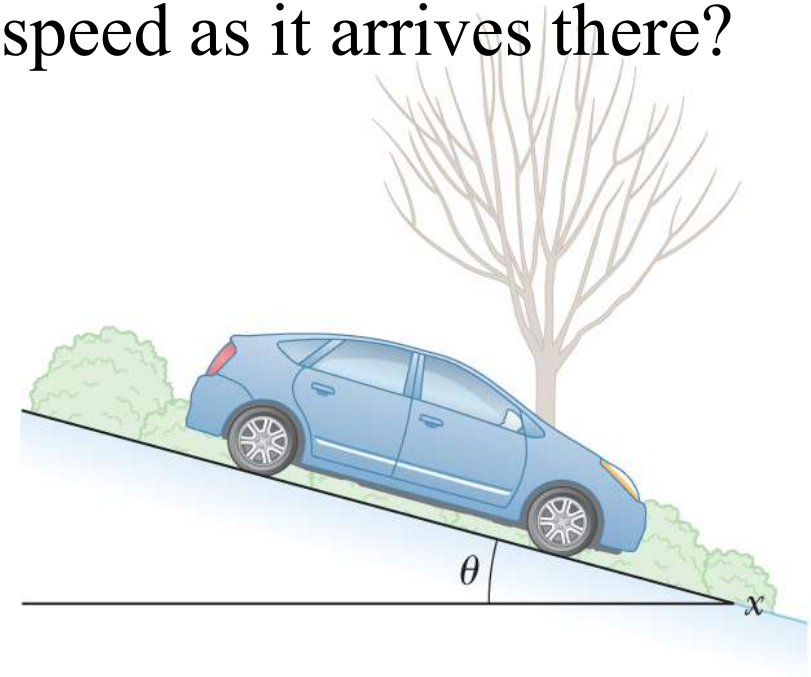
$$\sum F_y = n - mg\cos\theta = ma_y$$

$$a_x = g\sin\theta$$



Example 5.6:
The Runaway Car

(B) Suppose the car is released from rest at the top of the incline and the distance from the car's front bumper to the bottom of the incline is *d*. How long does it take the front bumper to reach the bottom of the hill, and what is the car's speed as it arrives there?



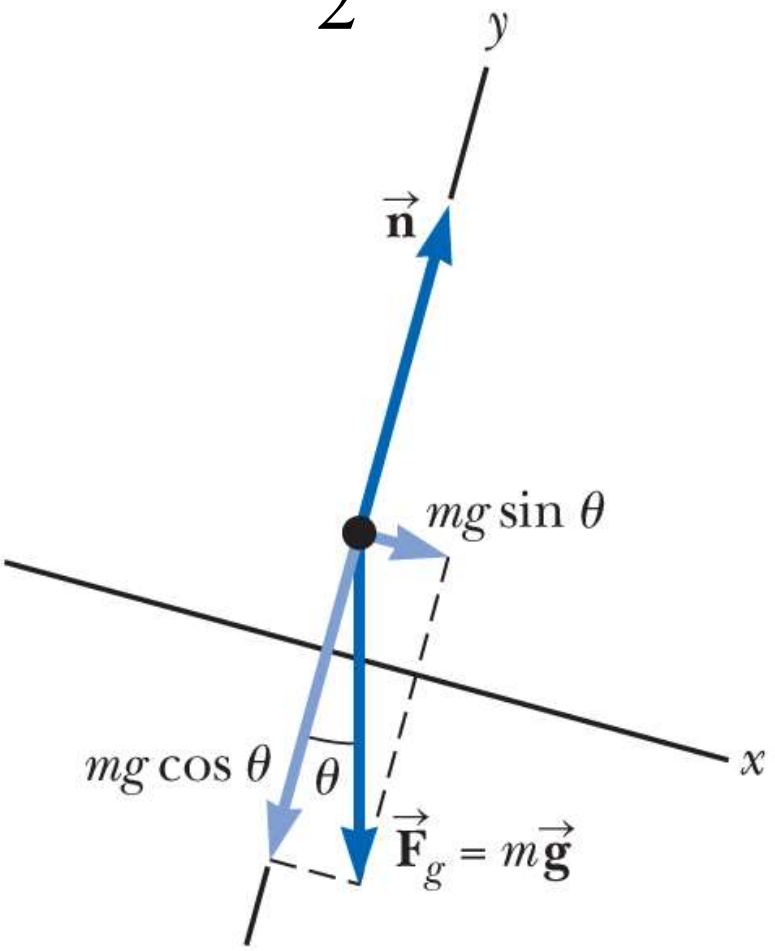
Example 5.6:
The Runaway Car

$$x_f = x_i + v_{xi}t + \frac{1}{2}a_x t^2 \rightarrow d = \frac{1}{2}a_x t^2$$

$$t = \sqrt{\frac{2d}{a_x}} = \boxed{\sqrt{\frac{2d}{g\sin\theta}}}$$

$$v_{xf}{}^2 = 2a_x d$$

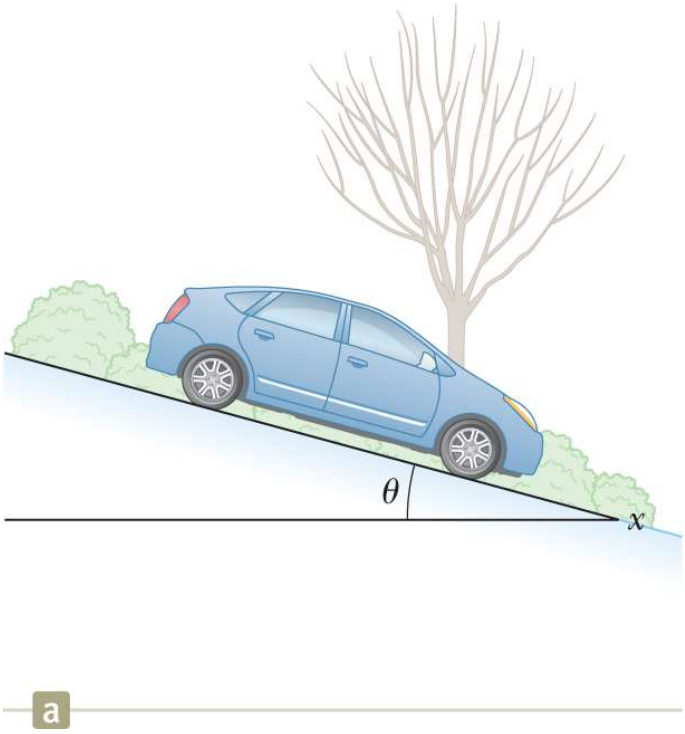
$$v_{xf} = \sqrt{2a_x d} = \boxed{\sqrt{2gd\sin\theta}}$$

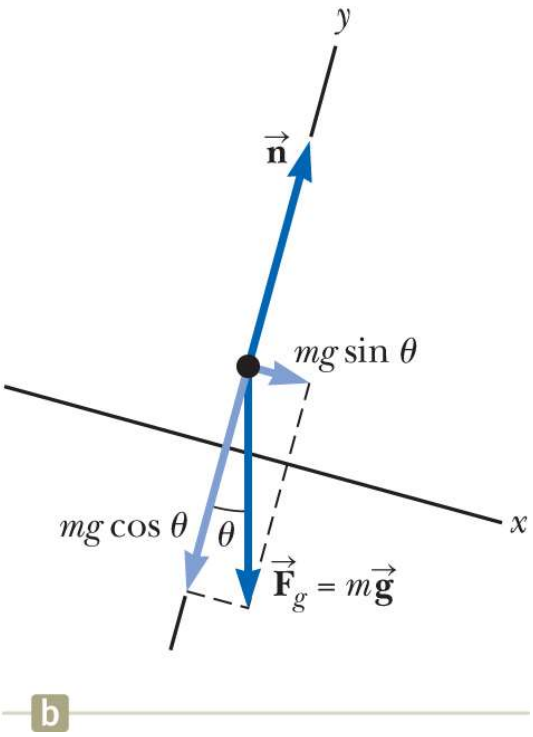


Example 5.6:
The Runaway Car

What previously solved problem does this situation become if $\theta = 90°$?

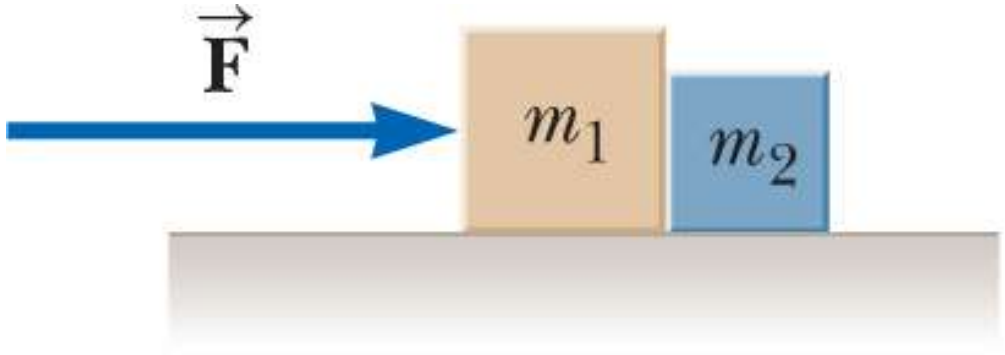
In free fall: $a_x = g\sin\theta = g\sin 90° = g$





Example 5.7:
One Block Pushes Another

Two blocks of masses m_1 and m_2, with $m_1 > m_2$, are placed in contact with each other on a frictionless, horizontal surface as in the figure. A constant horizontal force $\vec{\mathbf{F}}$ is applied to m_1 as shown.

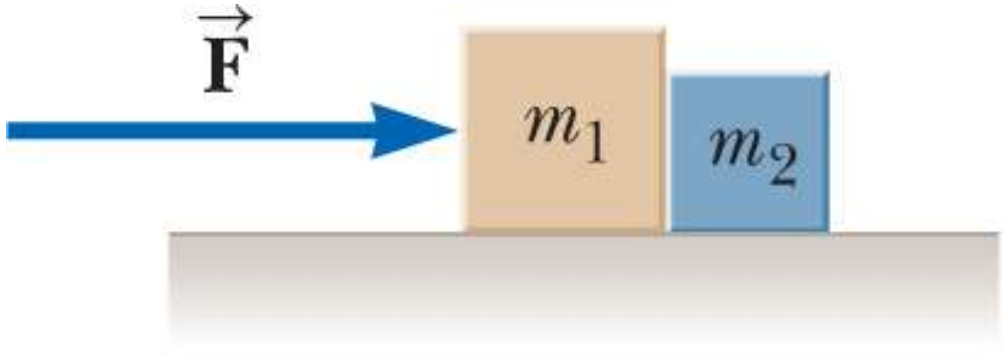


Example 5.7:
One Block Pushes Another

(A) Find the magnitude of the acceleration of the system.

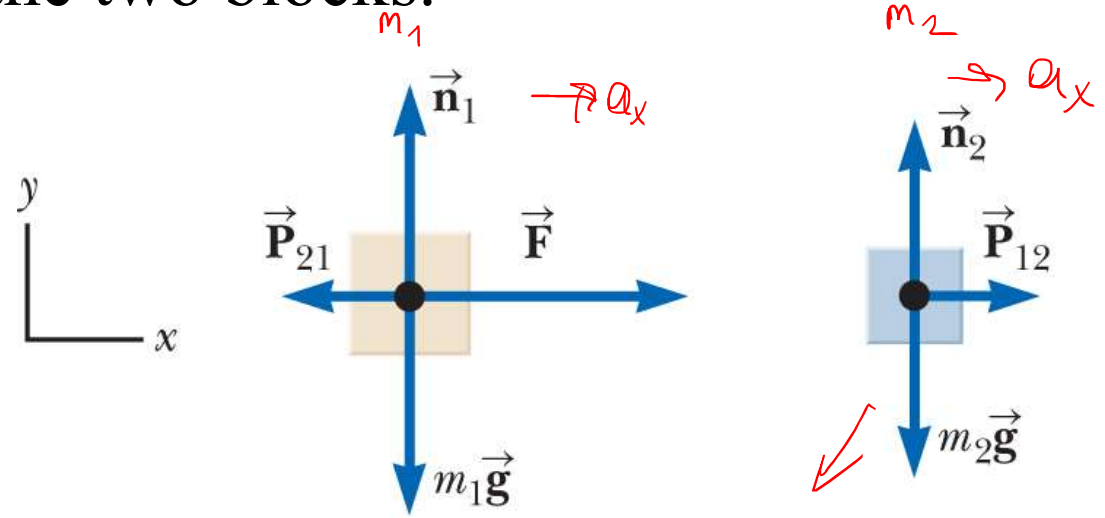
$$\sum F_x = F = \left(m_1 + m_2\right) a_x$$

$$a_x = \frac{F}{m_1 + m_2}$$



Example 5.7: One Block Pushes Another

(B) Determine the magnitude of the contact force between the two blocks.



$$\sum F_x = P_{12} = m_2 a_x$$

$$a_x = \frac{F}{m_1 + m_2} \rightarrow P_{12} = m_2 a_x = \left(\frac{m_2}{m_1 + m_2} \right) F$$

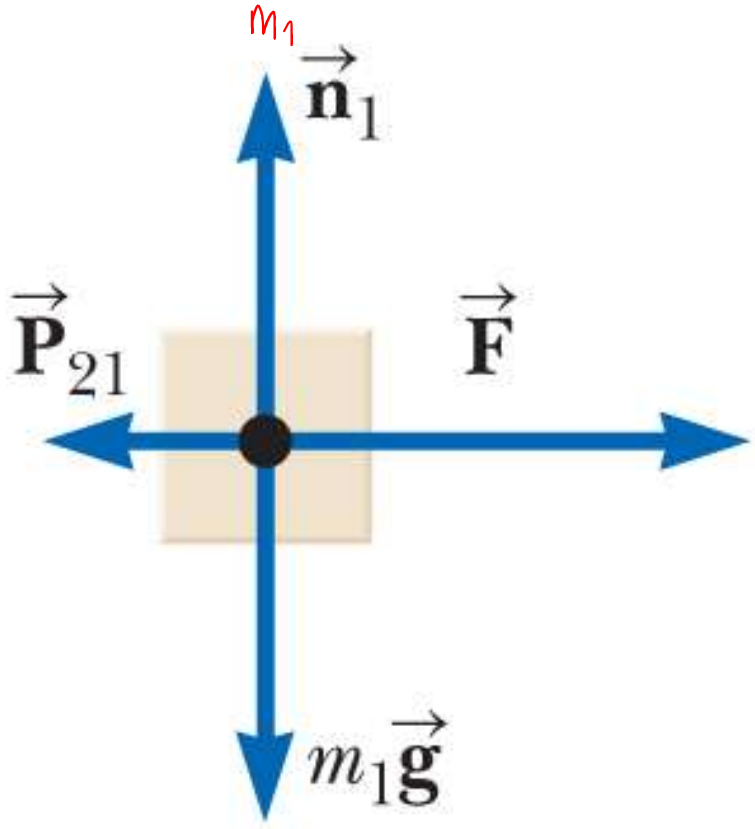
Example 5.7:
One Block Pushes Another

$$\sum F_x = F - P_{21} = F - P_{12} = m_1 a_x$$

$$P_{12} = F - m_1 a_x$$

$$= F - m_1 \left(\frac{F}{m_1 + m_2} \right)$$

$$P_{12} = \left(\frac{m_2}{m_1 + m_2} \right) F$$



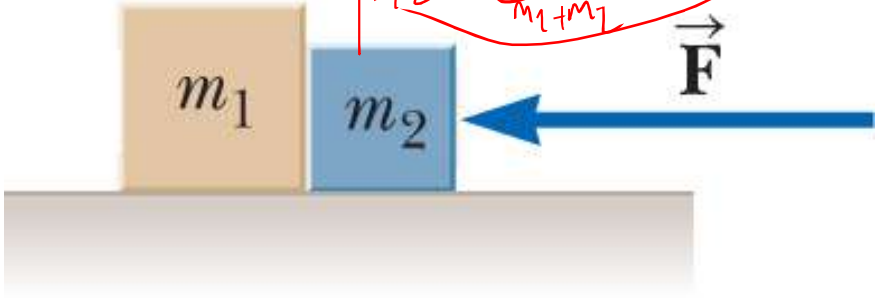
Example 5.7: One Block Pushes Another

Imagine that the force $\vec{\mathbf{F}}$ is is applied toward the left on the right-hand block of mass m_2. Is the magnitude of the force $\vec{\mathbf{P}}_{12}$ the same as it was when the force was applied toward the right on m_1?

When the force $\vec{\mathbf{F}}$ is applied from the right, we have:

$$\left|\vec{\mathbf{P}}_{12}\right| = \frac{m_1}{m_1 + m_2} F$$

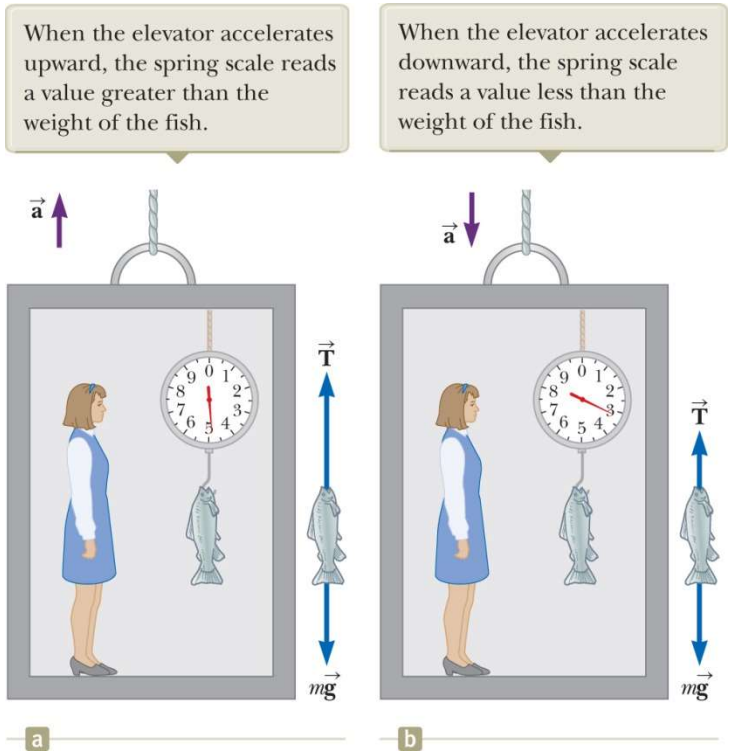
This is greater than before because $m_1 > m_2$.



Example 5.8:
Weighing a Fish in an Elevator

A person weighs a fish of mass *m* on a spring scale attached to the ceiling of an elevator as illustrated in the figure.

(A) Show that if the elevator accelerates either upward or downward, the spring scale gives a reading that is different from the weight of the fish.



Example 5.8: Weighing a Fish in an Elevator

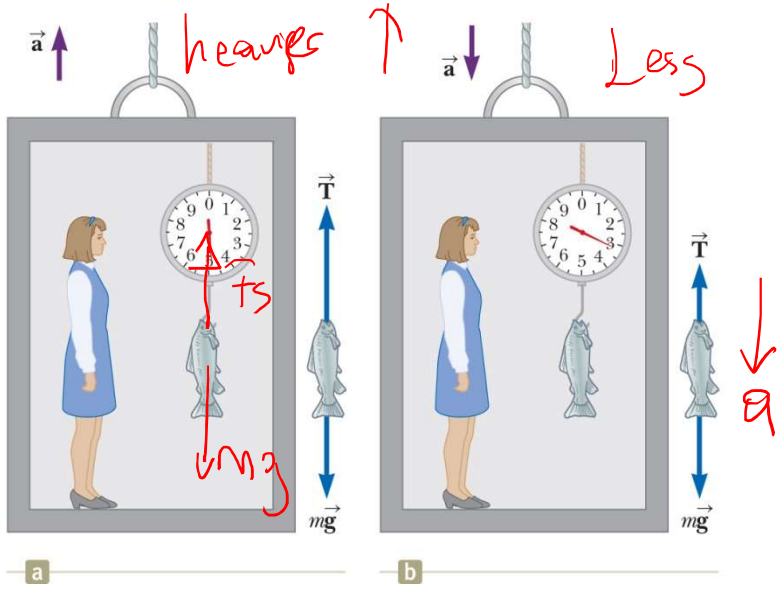
$$\sum F_y = T - F_g = 0 \Rightarrow T = F_g = mg$$

(elevator at rest or moving with constant v)

$$\sum F_y = T - mg = ma_y$$

$$T = ma_y + mg$$

$$= mg\left(\frac{a_y}{g} + 1\right) = F_g\left(\frac{a_y}{g} + 1\right)$$



Example 5.8: Weighing a Fish in an Elevator

(B) Evaluate the scale readings for a 40.0-N fish if the elevator moves with an acceleration $a_y = +2.00\ \text{m/s}^2$.

$$T = (40.0\ \text{N})\left(\frac{2.00\ \text{m/s}^2}{9.80\ \text{m/s}^2} + 1\right) = \boxed{48.2\ \text{N}}$$

$$T = (40.0\ \text{N})\left(\frac{-2.00\ \text{m/s}^2}{9.80\ \text{m/s}^2} + 1\right) = \boxed{31.8\ \text{N}}$$

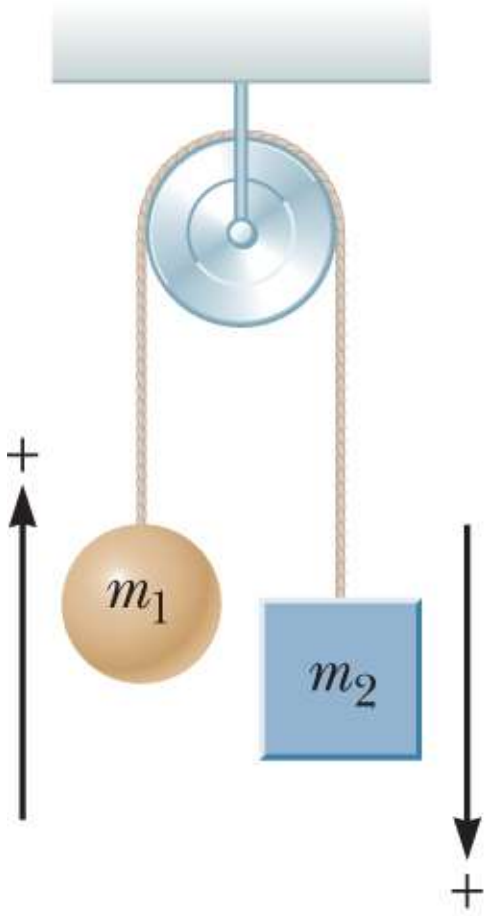
Example 5.8: Weighing a Fish in an Elevator

Suppose the woman in the figure tires of watching the scale and exits the elevator. Then the elevator cable breaks and the elevator and its remaining contents are in free fall. What happens to the reading on the scale?

$$T = F_g\left(\frac{a_y}{g}+1\right) \rightarrow T = F_g\left(\frac{-g}{g}+1\right) = 0$$

Example 5.9: The Atwood Machine

When two objects of unequal mass are hung vertically over a frictionless pulley of negligible mass as in the figure, the arrangement is called an *Atwood machine.* The device is sometimes used in the laboratory to determine the value of g by measuring the acceleration of the objects. Determine the magnitude of the acceleration of the two objects and the tension in the lightweight string.



Example 5.9: The Atwood Machine

$$\sum F_y = T - m_1 g = m_1 a_y \quad (1)$$

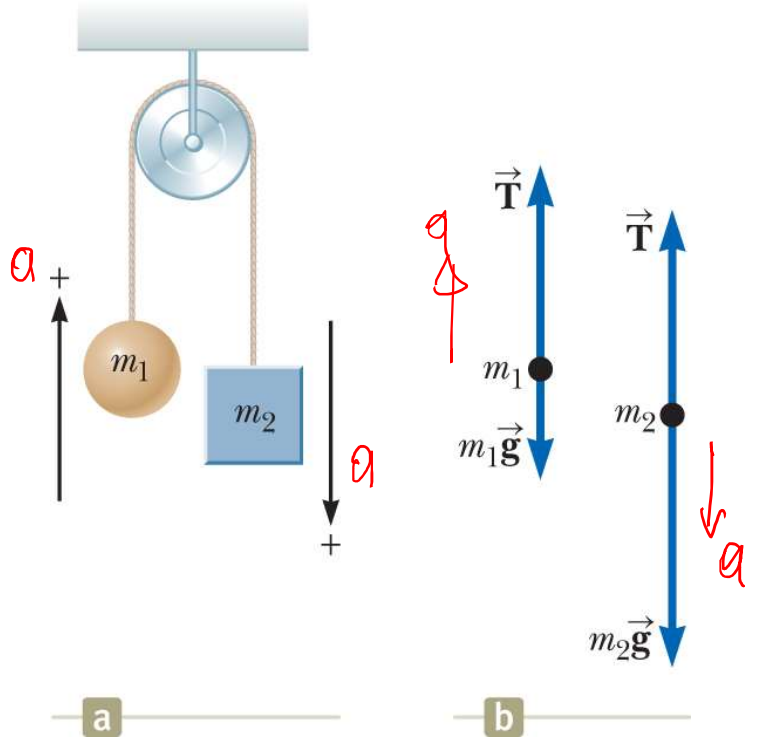
$$\sum F_y = m_2 g - T = m_2 a_y \quad (2)$$

(1)+(2) →

$$-m_1 g + m_2 g = m_1 a_y + m_2 a_y$$

$$a_y = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) g$$

$$T = m_1 \left(g + a_y \right) = \left(\frac{2 m_1 m_2}{m_1 + m_2} \right) g$$



Example 5.9:
The Atwood Machine

Describe the motion of the system if the objects have equal masses, that is, $m_1 = m_2$.

No acceleration

for $m_1 = m_2 = m$:

$$a_y = \left(\frac{m-m}{m+m}\right)g \Rightarrow a_y = \left(\frac{0}{2m}\right)g = 0$$

Example 5.9: The Atwood Machine

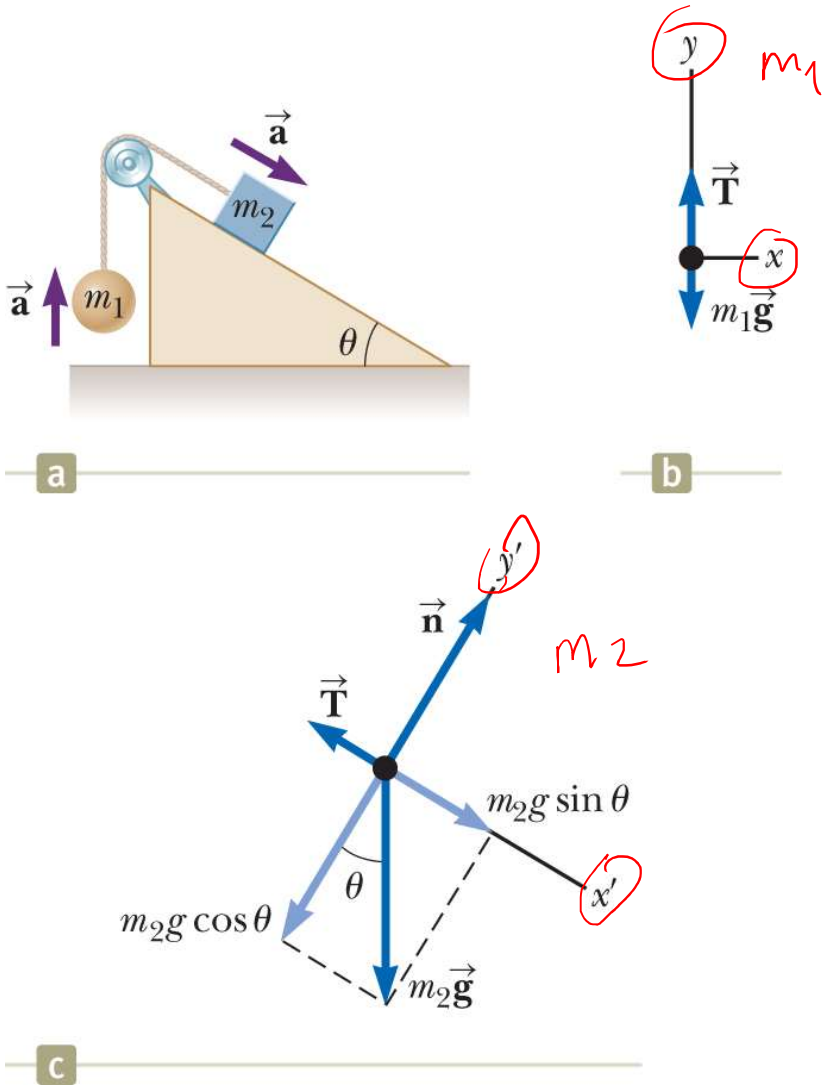
What if one of the masses is much larger than the other: $m_1 >> m_2$?

m_1 falls as if m_2 were not there

$$\text{if } m_1 >> m_2 \rightarrow a_y = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) g \approx \left(\frac{-m_1}{m_1}\right) g = -g$$

Example 5.10: Acceleration of Two Objects Connected by a Cord

A ball of mass m_1 and a block of mass m_2 are attached by a lightweight cord that passes over a frictionless pulley of negligible mass as in the top figure. The block lies on a frictionless incline of angle θ. Find the magnitude of the acceleration of the two objects and the tension in the cord.



Example 5.10: Acceleration of Two Objects Connected by a Cord

$$\sum F_y = T - m_1 g = m_1 a_y = m_1 a \quad (1)$$

$$\sum F_{x'} = m_2 g \sin\theta - T = m_2 a_{x'} = m_2 a \quad (2)$$

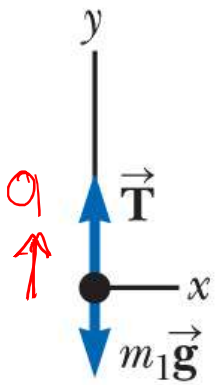
$$\sum F_{y'} = n - m_2 g \cos\theta = 0 \quad (3)$$

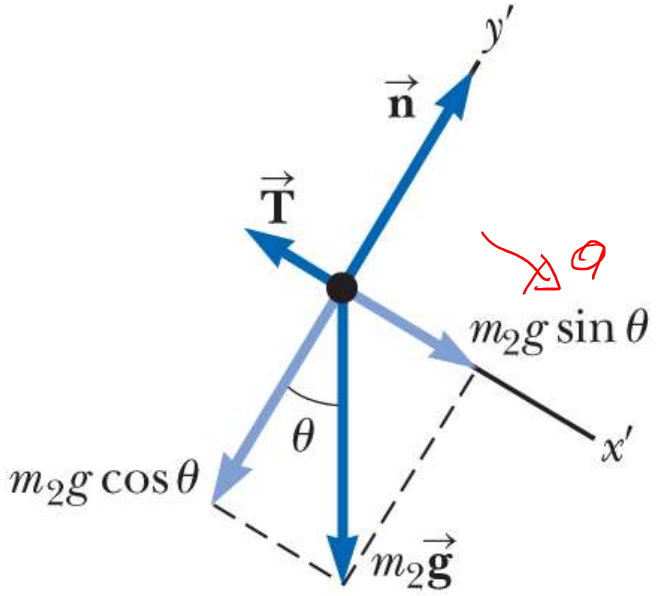
$$T = m_1 (g + a)$$

$$m_2 g \sin\theta - m_1 (g + a) = m_2 a$$

$$a = \left(\frac{m_2 \sin\theta - m_1}{m_1 + m_2} \right) g$$

$$T = \left(\frac{m_1 m_2 (\sin\theta + 1)}{m_1 + m_2} \right) g$$





Example 5.10: Acceleration of Two Objects Connected by a Cord

What happens in this situation if $\theta = 90°$?

It becomes an Atwood machine!

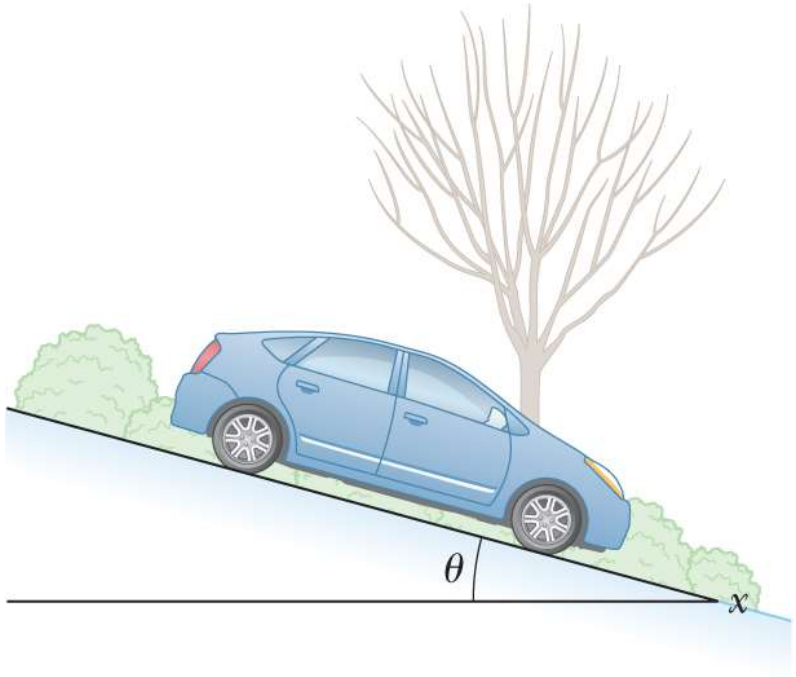
$$a = \left(\frac{m_2 \sin\theta - m_1}{m_1 + m_2} \right) g \rightarrow a = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) g$$

$$T = \left(\frac{m_1 m_2 \left(\sin\theta + 1 \right)}{m_1 + m_2} \right) g \rightarrow T = \left(\frac{2 m_1 m_2}{m_1 + m_2} \right) g$$

Example 5.10: Acceleration of Two Objects Connected by a Cord

What if $m_1 = 0$?

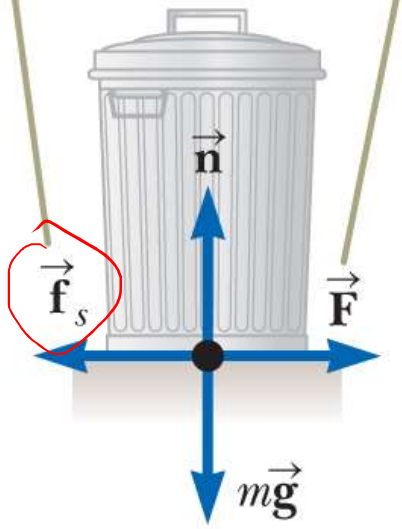
Describes a mass sliding down a frictionless inclined plane, similar to the sliding car problem



$$a = \left(\frac{m_2 \sin\theta - m_1}{m_1 + m_2} \right) g \rightarrow a = \left(\frac{m_2 \sin\theta - 0}{0 + m_2} \right) g = \left(\frac{m_2 \sin\theta}{m_2} \right) g$$

Forces of Friction

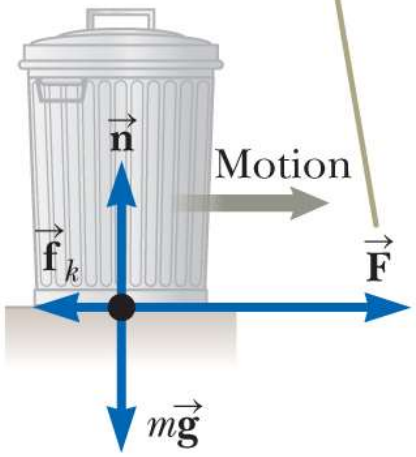
For small applied forces, the magnitude of the force of static friction equals the magnitude of the applied force.

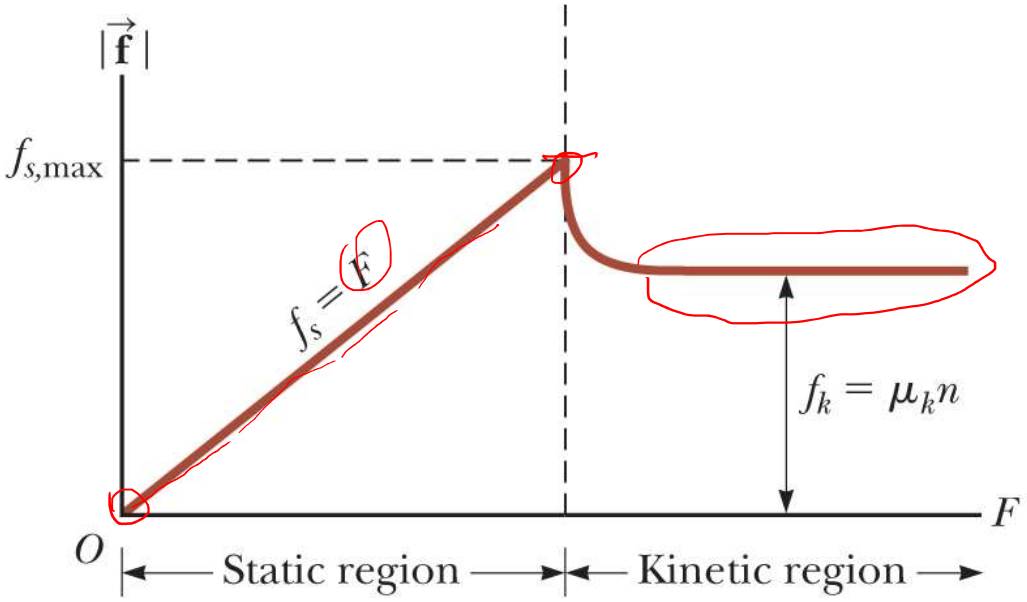


$f_s \rightarrow$ static friction force

Forces of Friction

When the magnitude of the applied force exceeds the magnitude of the maximum force of static friction, the trash can breaks free and accelerates to the right.





Coefficients of Friction



μ_s and μ_k are independent of area.

$$f_s \leq \mu_s n$$

μ_s: coefficient of static friction

$$f_s = f_{s,\max} = \mu_s n$$

$$f_k = \mu_k n$$

μ_k: coefficient of kinetic friction

TABLE 5.1 Coefficients of Friction

	μ_s	μ_k
Rubber on concrete	1.0	0.8
Steel on steel	0.74	0.57
Aluminum on steel	0.61	0.47
Glass on glass	0.94	0.4
Copper on steel	0.53	0.36
Wood on wood	0.25–0.5	0.2
Waxed wood on wet snow	0.14	0.1
Waxed wood on dry snow	—	0.04
Metal on metal (lubricated)	0.15	0.06
Teflon on Teflon	0.04	0.04
Ice on ice	0.1	0.03
Synovial joints in humans	0.01	0.003

Note: All values are approximate. In some cases, the coefficient of friction can exceed 1.0.

Quick Quiz 5.6

You press your physics textbook flat against a vertical wall with your hand. What is the direction of the friction force exerted by the wall on the book?

(a) downward

(b) upward

(c) out from the wall

(d) into the wall

Quick Quiz 5.6

You press your physics textbook flat against a vertical wall with your hand. What is the direction of the friction force exerted by the wall on the book?

(a) downward

(b) upward

(c) out from the wall

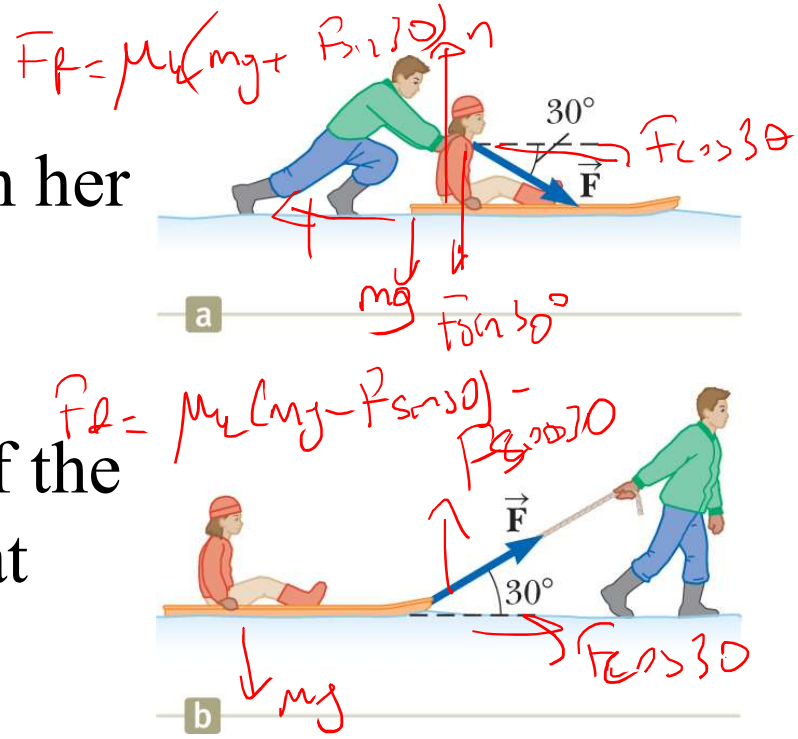
(d) into the wall

Quick Quiz 5.7

Charlie is playing with his daughter Torrey in the snow. She sits on a sled and asks him to slide her across a flat, horizontal field. Charlie has a choice of

(a) pushing her from behind by applying a force downward on her shoulders at 30° below the horizontal or

(b) attaching a rope to the front of the sled and pulling with a force at 30° above the horizontal.

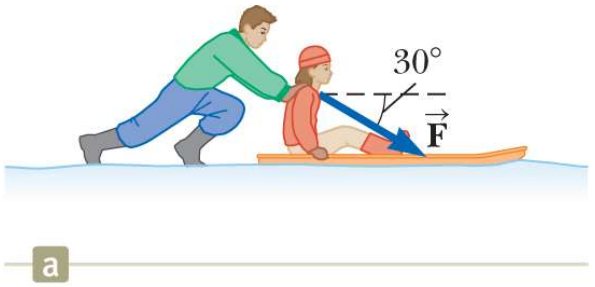


Which would be easier for him and why?

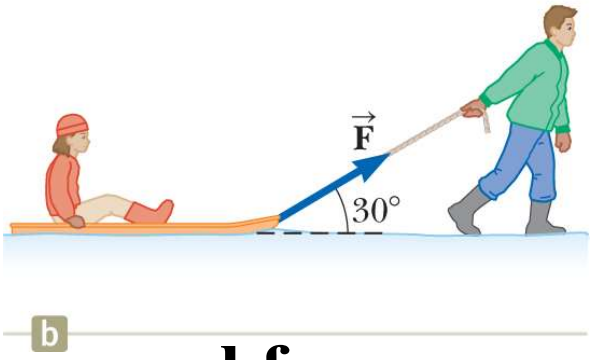
Quick Quiz 5.7

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(a) pushing her from behind by applying a force downward on her shoulders at 30° below the horizontal or



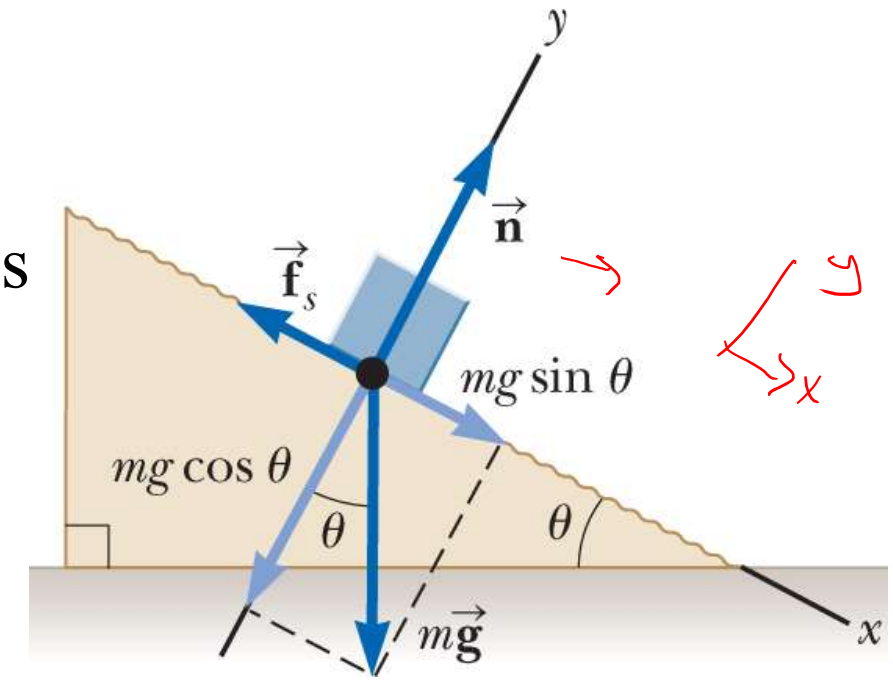
(b) attaching a rope to the front of the sled and pulling with a force at 30° above the horizontal.



Pulling up on the rope decreases the normal force, which, in turn, decreases the force of kinetic friction.

Example 5.11: Experimental Determination of μ_s and μ_k

The following is a simple method of measuring coefficients of friction. Suppose a block is placed on a rough surface inclined relative to the horizontal as shown in the figure. The incline angle is increased until the block starts to move. Show that you can obtain μ_s by measuring the critical angle θ_c at which this slipping just occurs.



Example 5.11: Experimental Determination of μ_s and μ_k

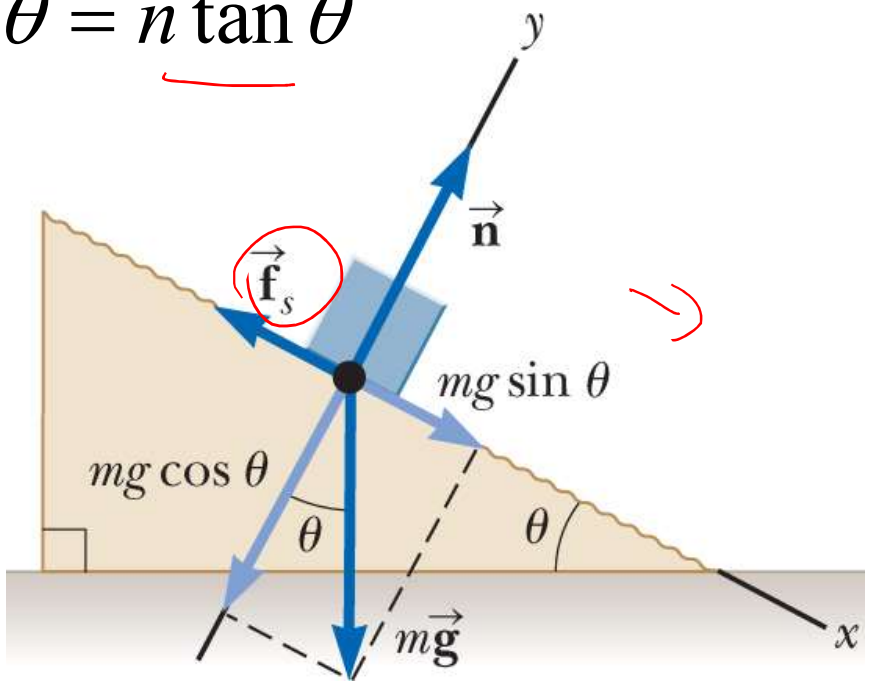
$$\sum F_x = mg\sin\theta - f_s = 0$$

$$\sum F_y = n - mg\cos\theta = 0$$

$$f_s = mg\sin\theta = \left(\frac{n}{\cos\theta}\right)\sin\theta = n\tan\theta$$

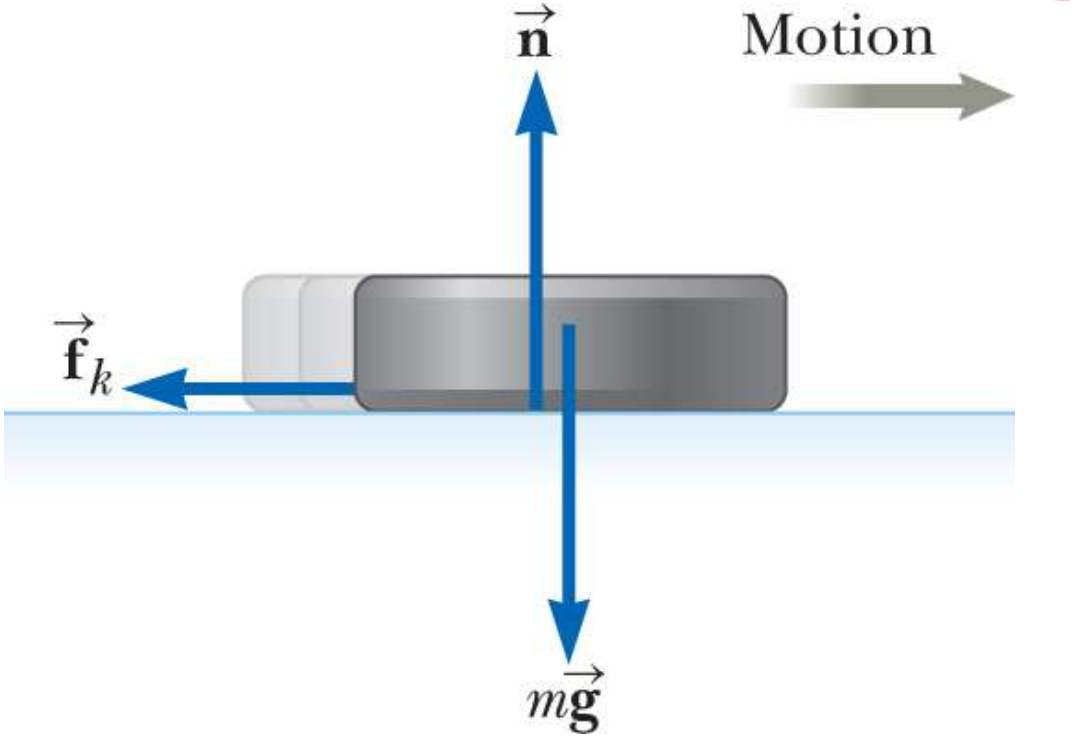
$$\mu_s n = n\tan\theta_c$$

$$\mu_s = \tan\theta_c$$



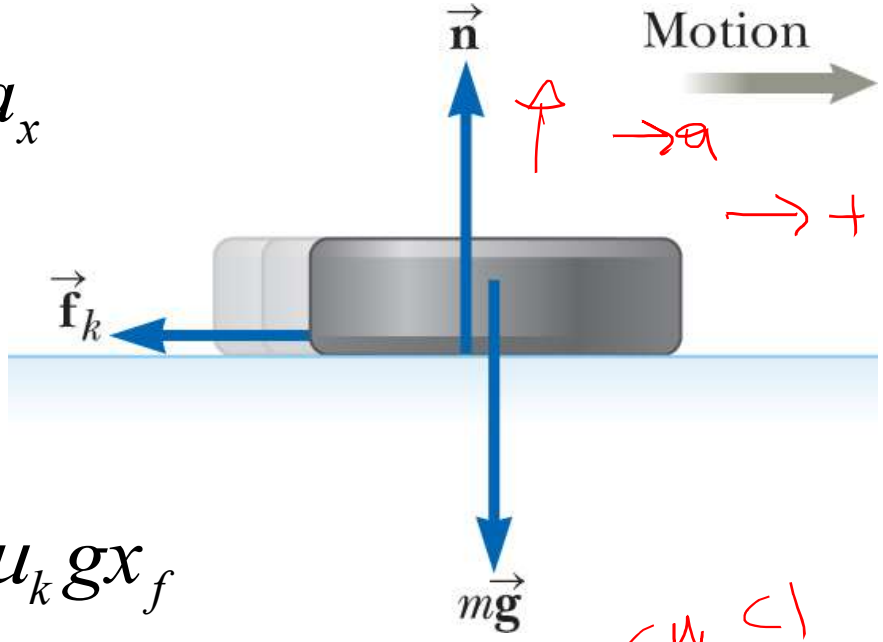
Example 5.12:
The Sliding Hockey Puck

A hockey puck on a frozen pond is given an initial speed of 20.0 m/s. If the puck always remains on the ice and slides 115 m before coming to rest, determine the coefficient of kinetic friction between the puck and ice.



Example 5.12: The Sliding Hockey Puck

$$\sum F_x = -f_k = ma_x \qquad \sum F_y = n - mg = 0$$



$$-\mu_k n = -\mu_k mg = ma_x$$

$$a_x = -\mu_k g$$

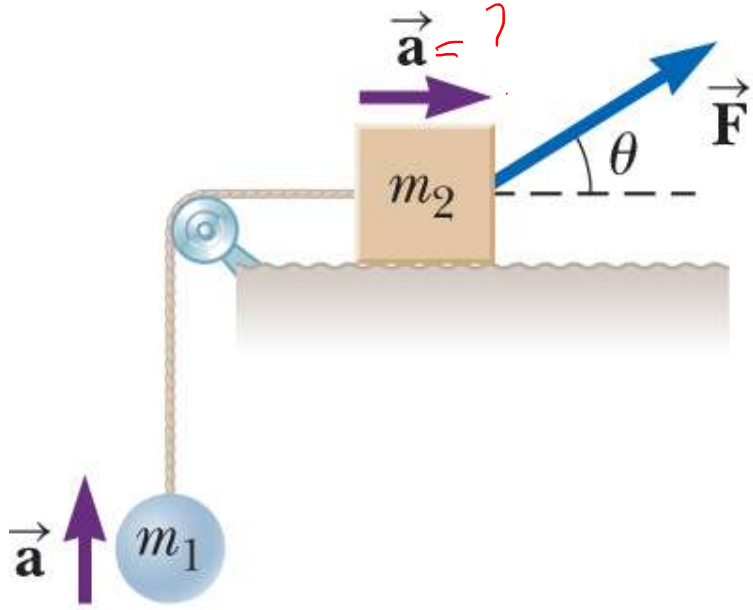
$$v_{xf}^{\,2} = v_{xi}^{\,2} + 2a_x\left(x_f - x_i\right)$$

$$0 = v_{xi}^{\,2} + 2a_x x_f = v_{xi}^{\,2} - 2\mu_k g x_f$$

$$\mu_k = \frac{v_{xi}^{\,2}}{2gx_f} \qquad \mu_k = \frac{\left(20.0\ \text{m/s}\right)^2}{2\left(9.80\ \text{m/s}^2\right)\left(115\ \text{m}\right)} = \boxed{0.177}$$

Example 5.13: Acceleration of Two Connected Objects When Friction Is Present

A block of mass m_2 on a rough, horizontal surface is connected to a ball of mass m_1 by a lightweight cord over a lightweight, frictionless pulley as shown in the figure. A force of magnitude F at an angle θ with the horizontal is applied to the block as shown, and the block slides to the right. The coefficient of kinetic friction between the block and surface is μ_k. Determine the magnitude of the acceleration of the two objects.



Example 5.13: Acceleration of Two Connected Objects When Friction Is Present

$$\sum F_x = F\cos\theta - f_k - T = m_2 a_x = m_2 a \quad (1)$$

$$\sum F_y = n + F\sin\theta - m_2 g = 0 \quad (2)$$

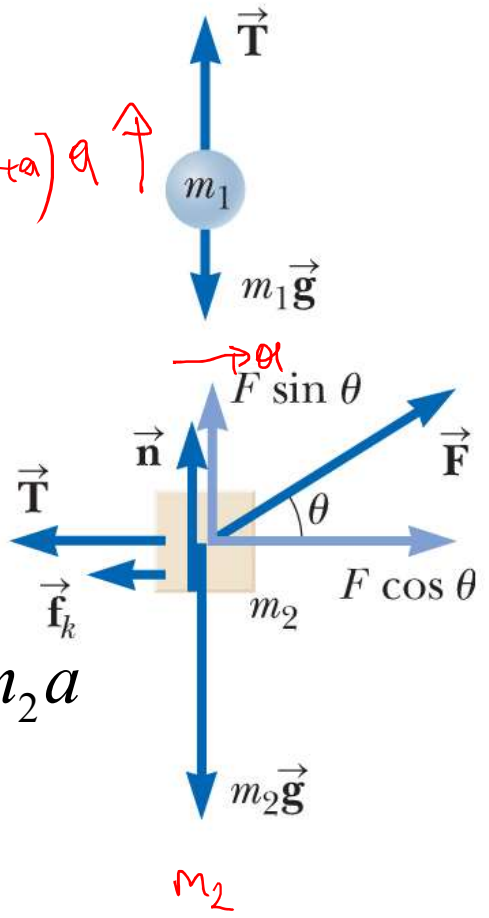
$$\sum F_y = T - m_1 g = m_1 a_y = m_1 a \quad (3)$$

$$n = m_2 g - F\sin\theta$$

$$f_k = \mu_k\left(m_2 g - F\sin\theta\right)$$

$$F\cos\theta - \mu_k\left(m_2 g - F\sin\theta\right) - m_1\left(a + g\right) = m_2 a$$

$$a = \frac{F\left(\cos\theta + \mu_k \sin\theta\right) - \left(m_1 + \mu_k m_2\right) g}{m_1 + m_2}$$



Assessing to Learn

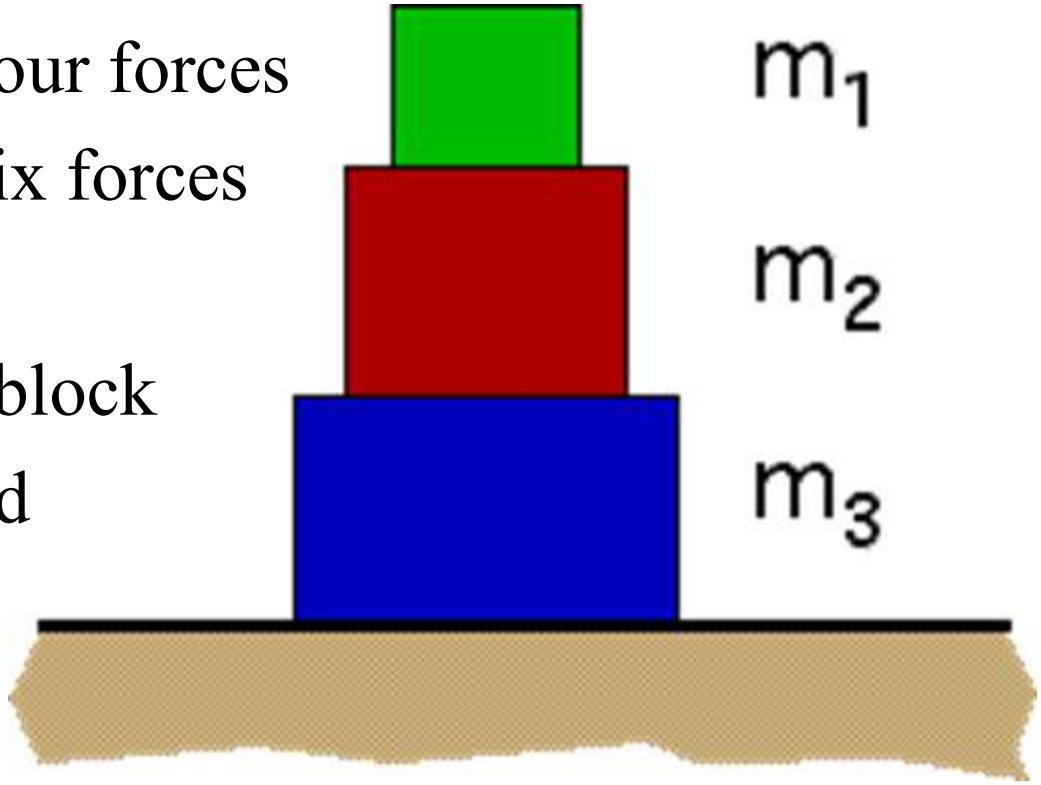
A baseball is struck by a bat. While the ball is in the air, what objects exert forces on the ball?

1. Earth
2. Bat
3. Air
4. Bat, Air
5. Earth, Bat
6. Earth, Air
7. Earth, Bat, Air
8. There are no forces on the ball.
9. The answer depends on whether the ball is going up, going down, or at its highest point.

Assessing to Learn

Three blocks are stacked as shown below.
How many forces are acting on the bottom block (m_3)?

1. One force
2. Two forces
3. Three forces
4. Four forces
5. Five forces
6. Six forces
7. More than six forces
8. No forces act on the block
9. Cannot be determined



Assessing to Learn

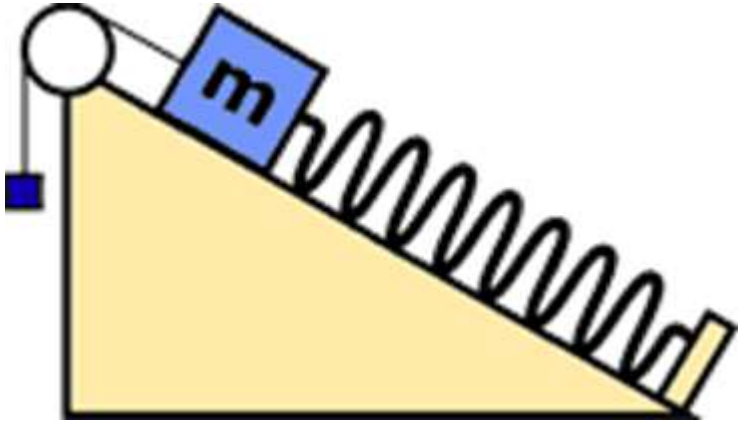
A block of mass *m* is on a rough surface, with a spring attached and extended. As the block moves up the incline a small distance, how many forces are exerted on the mass?

1. One force 2. Two forces 3. Three forces 4. Four forces 5. Five forces 6. Six forces

7. Seven forces 8. More than 7 forces

9. None of the above

9. Impossible to determine



Assessing to Learn

A monkey hangs on a rope. What forces act on the monkey? (Ignore forces due to the air.)

1. Friction, Gravitation
2. Tension, Gravitation
3. Friction, Tension, Gravitation
4. Normal, Friction, Gravitation
5. More than one answer is true
6. None of the above
7. Cannot be determined

Assessing to Learn

A thin wire is stretched horizontally between two walls. If a weight W is hung on the wire, what is true about the tension T in the wire?

1. $T < W$
2. $T = W$
3. $T > W$
4. The relationship between T and W cannot be determined.

Assessing to Learn

Consider the following three items.
A. 100 g of steam on the Moon
B. 10 g of water on the Earth
C. 20 g of ice floating in water on the Earth

Put these items in order of increasing **weight**:

1. A < B < C
2. A < C < B
3. B < A < C
4. B < C < A
5. C < A < B
6. C < B < A
7. None of the above
8. Impossible to determine

Assessing to Learn

Consider the following three items.
A. 100 g of steam on the Moon
B. 10 g of water on the Earth
C. 20 g of ice floating in water on the Earth

Put these items in order of increasing **mass**:

1. A < B < C
2. A < C < B
3. B < A < C
4. B < C < A
5. C < A < B
6. C < B < A
7. None of the above
8. Impossible to determine

Assessing to Learn

Consider the following three items.
A. 100 g of steam on the Moon
B. 10 g of water on the Earth
C. 20 g of ice floating in water on the Earth

Put these items in order of increasing **density**:

1. A < B < C
2. A < C < B
3. B < A < C
4. B < C < A
5. C < A < B
6. C < B < A
7. None of the above
8. Impossible to determine

Assessing to Learn

An astronaut floats inside an orbiting space station.
Which of the following are true?

A. No forces act on the astronaut.
B. The astronaut has no mass.
C. The astronaut has no weight.

1. A only 2. B only 3. C only

3. A and B 5. A and C 6. B and C

7. all are true

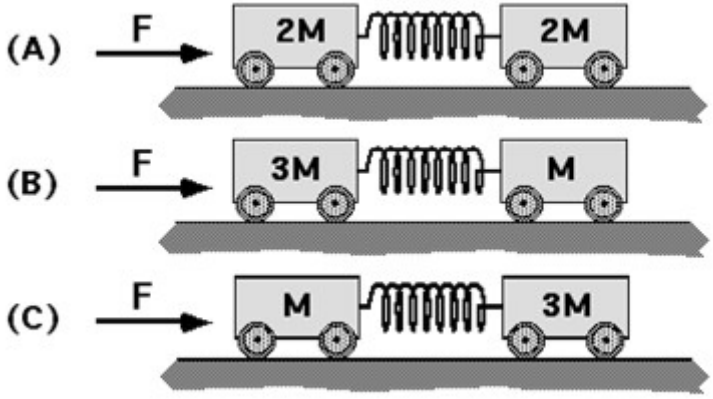
8. none are true

Assessing to Learn

Consider the three situations shown below. In each case two small carts are connected by a spring. A constant force *F* is applied to the leftmost cart in each case. In each situation the springs are compressed so that the distance between the two carts never changes.

Which of the following statements must be true regarding the compression of the spring in each case? Assume the springs are identical.

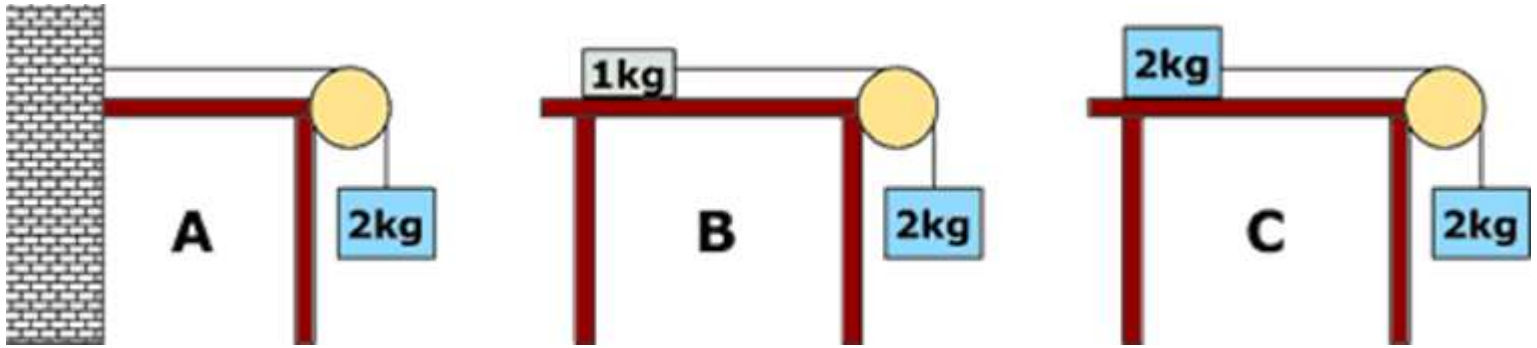
1. Compression A = Compression B = Compression C
2. B = C < A
3. A < B = C
4. A < B < C
5. B < A < C
6. C < A < B
7. A < C < B
8. None of the above
9. Cannot be determined



Assessing to Learn

Consider the three situations below, labeled A, B, and C. Ignore friction. After each system is released from rest, how do the tensions in the strings compare?

1. A = B = C
2. B = C < A
3. A = C < B
4. A < B < C
5. A < C < B
6. B < A < C
7. B < C < A
8. C < A < B
9. C < B < A
10. Impossible to determine



Assessing to Learn

Two blocks are arranged as shown and kept at rest by holding the 1 kg block in place. The tension in the string is **closest** to:

1. 9 N 2. 10 N 3. 11 N 4. 12 N 5. 13 N

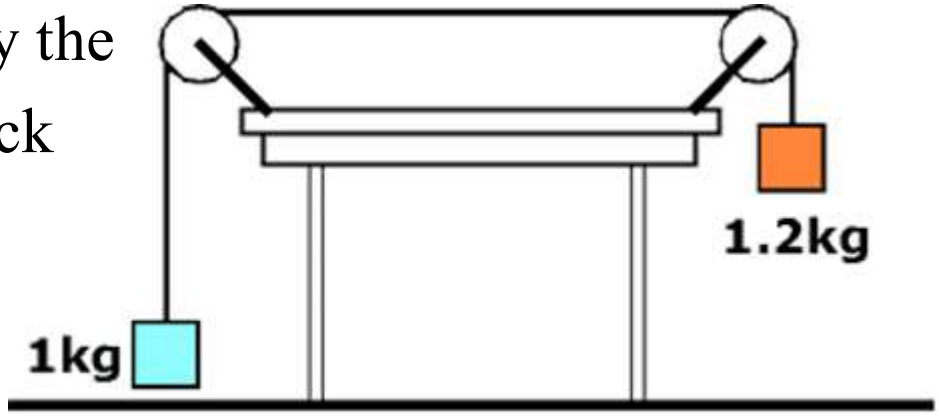
6. 10 N at the left end; 12 N at the right

7. 10 N in the left segment; 11 N in the middle segment; 12 N in the right segment

8. Smoothly varying from 10 N by the left block to 12 N by the right block

9. None of the above

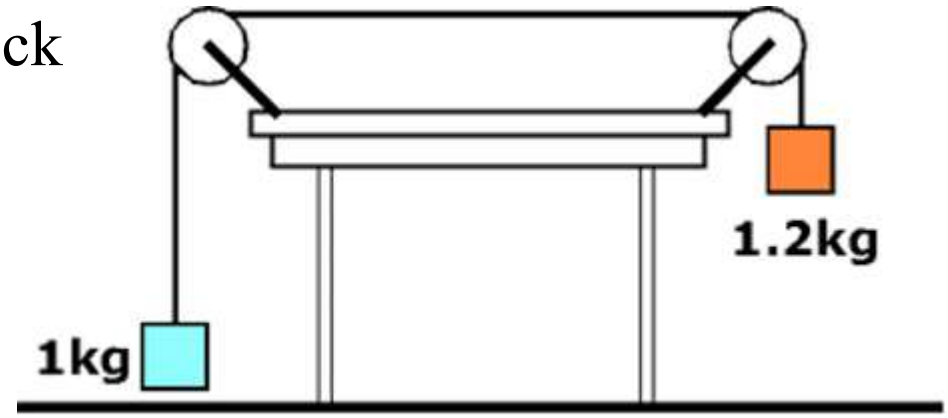
10. Impossible to determine



Assessing to Learn

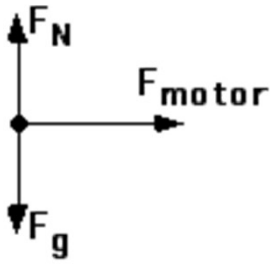
Two blocks are arranged as shown and released from rest. The tension in the string is **closest** to:

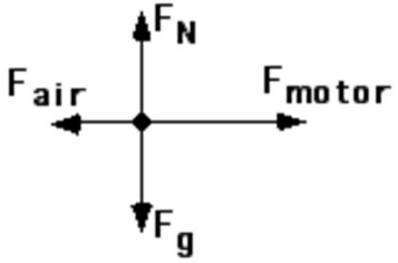
1. 9 N 2. 10 N 3. 11 N 4. 12 N 5. 13 N
6. 10 N at the left end; 12 N at the right
7. 10 N in the left segment; 11 N in the middle segment; 12 N in the right segment
8. Smoothly varying from 10 N by the left block to 12 N by the right block
9. None of the above
10. Impossible to determine

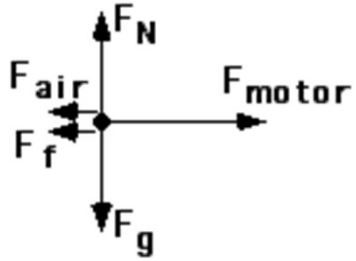


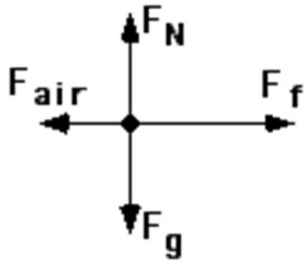
Assessing to Learn

A car accelerates down a straight highway. Which of the free-body diagrams shown below best represents the forces on the car?

1. 



3. ‘ 

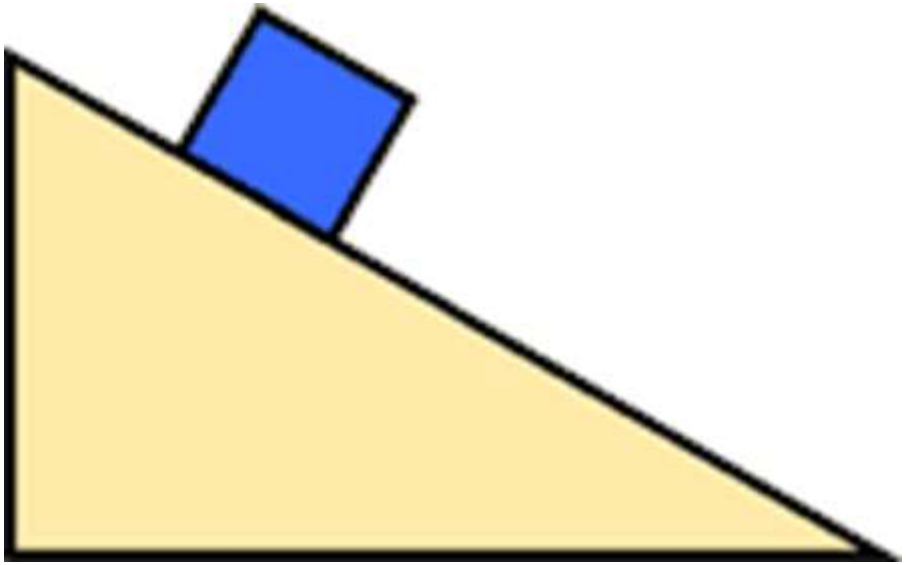


5. None of these 6. Cannot be determined

Assessing to Learn

A mass of 5 kg sits at rest on an incline making an angle of 30° to the horizontal. If $\mu_s = 0.7$, what is the friction force on the block?

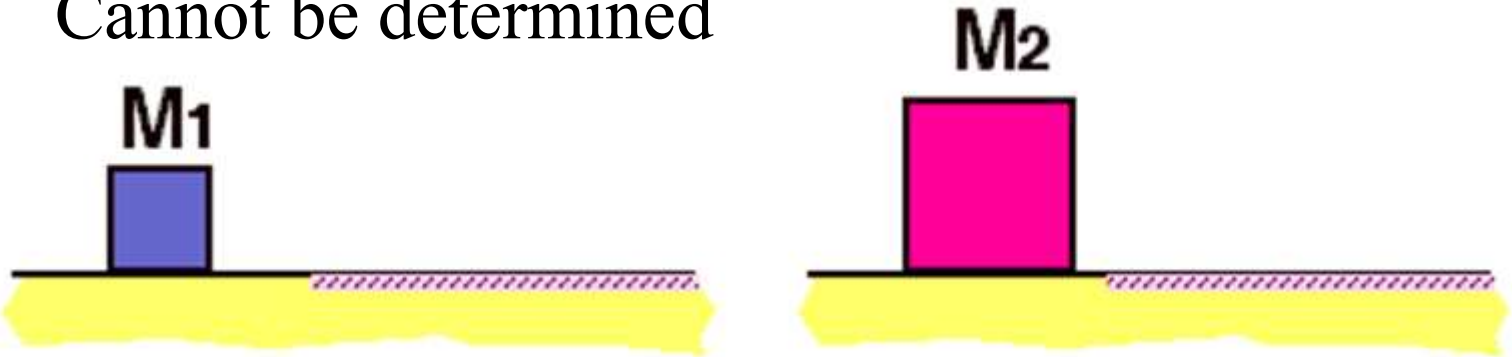
1. 43.3 N, down the incline
2. 25 N, up the incline
3. 10 N, down the incline
4. 30.3 N, up the incline
5. None of the above



Assessing to Learn

Two blocks, having the same mass but different sizes, slide with the same constant speed on a smooth surface, then move onto a surface having friction coefficient μ_k. Which stops in the shorter time?

1. M_1
2. M_2
3. Both stop in the same time
4. Cannot be determined



Assessing to Learn

Two blocks, $M_1 > M_2$, having the same speed, move from a frictionless surface onto a surface having friction coefficient μ_k. Which stops in the shorter time?

1. M_1
2. M_2
3. Both stop in the same time
4. Cannot be determined

